

A two-coordinate obstruction for affine interpretations of the Yolcu–Aaronson–Heule Collatz system

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Abstract

We prove two obstructions to matrix-interpretation proofs for the eleven-rule Collatz system of Yolcu, Aaronson, and Heule. For nonnegative real affine maps in two coordinates, if every symbol matrix has first diagonal entry at least one, weak orientation of all eleven rules forces every first-coordinate offset gap to vanish. This holds separately for the original and word-reversed systems, with arbitrary admissible boundary maps and no coefficient bound. For the reversed system we also remove every diagonal lower bound: nonnegativity and the eleven weak comparisons alone force equality of the offset vectors in both rules eligible for the reversed TOP step, $da \rightarrow d$ and $db \rightarrow dg$. Thus neither rule can have a positive uniform decrement in this two-coordinate real affine format. The proofs include singular matrices and zero entries. We give the matrix and scalar classifications, the resonance calculation, and the growth argument underlying the unrestricted reversed result. Lean 4 checks both theorems and their supporting algebra. The forward TOP problem, higher dimensions, and the Collatz conjecture remain unresolved by these results.

1 Introduction

Yolcu, Aaronson, and Heule constructed a mixed binary–ternary string rewriting system whose termination is equivalent to the Collatz conjecture [1, Theorem 3.17]. Their construction turns a question about integer trajectories into a termination problem for eleven rules. The system, its arithmetic interpretation, and that equivalence are their results. This paper studies a restriction on possible matrix-interpretation proofs for their system.

Rule removal builds a termination proof in stages. An interpretation orients all remaining rules weakly and a nonempty subset strictly; relative termination then permits removal of that subset. Repeating the process can establish termination [1, Theorems 2.6 and 2.15]. The first step therefore needs at least one strict rule while retaining weak comparisons for the complete system.

Our first result rules out that first step in two coordinates for the strictly monotone affine format defined below. It quantifies over all admissible nonnegative real coefficients, including irrational ones, and imposes no upper bound. A failed bounded SAT search cannot establish this conclusion: it leaves larger coefficients and coefficients outside its encoding unexamined. The obstruction is algebraic. It forces all eleven candidate strict gaps to vanish, rather than excluding one chosen rule or one coefficient range.

Dimension restrictions matter in this setting. Yolcu, Aaronson, and Heule report that, for some weakened variants, their tool found only interpretations of dimension five [1, Section 1]. Their Table 1 also records a natural dimension-five certificate for the subsystem omitting $bf \rightarrow ga$ in our notation. These are search outcomes, not proofs of minimal dimension. The present result supplies a coefficient-independent exclusion for the full system in dimension two, within a narrower, precisely stated interpretation format.

The proof has two parts. First, the matrix comparisons force the five digit matrices to share a triangular orientation, with unit first diagonals. Second, an affine analysis forces every first-offset gap to zero. The upper triangular case requires retaining products between matrix entries and

offsets, including several zero-slope cases. A second result removes the diagonal bounds for the reversed TOP problem. Assuming a strict eligible rule, it first forces a common triangular orientation. It then excludes each possible active diagonal, using a comparison between binary-prefix growth and ternary-word bounds in the expanding case. All obstruction results are formalized in Lean. Appendix A reproduces the definitions and both exact final theorem types.

1.1 Related work

Matrix-interpretation obstructions for Collatz rewriting already appear in Yolcu, Aaronson, and Heule’s paper. Their starting point is Zantema’s seven-rule unary system, whose termination is equivalent to the Collatz conjecture [5, Section 10, Theorem 16]. For that system, denoted Z in [1], Theorem 3.8 of [1] excludes natural matrix interpretations in every finite dimension that weakly orient all rules and strictly orient at least one, under extended monotonicity. Thus no direct rule-removal step is possible in that format.

Their dependency-pair result is also an impossibility theorem. After other pairs have been removed, Lemma 3.9 reduces the problem to two residual pairs. Theorem 3.10 excludes a natural matrix interpretation, in any dimension, that strictly orients either residual pair while weakly orienting Z , even with only weak monotonicity. Both obstructions also hold for the reversed unary system [1, Section 3.1.1]. The dependency-pair statement therefore allows the earlier removals; it identifies the obstruction that remains afterward.

Our results address the eleven-rule mixed binary–ternary system instead. Both quantify over nonnegative real coefficients, including irrational ones, in dimension two. Theorem 1 retains the diagonal bound in (2) and makes all eleven first-offset gaps zero in either word orientation.

Theorem 2 removes every diagonal lower bound for the reversed system. Under its eleven weak comparisons, the offset vectors of da and d coincide, as do those of db and dg . Consequently neither of these two rules admits a uniform positive decrement in either output coordinate. These are exactly the eligible reversed dynamic rules in the relative top-removal reduction of Yolcu–Aaronson–Heule [1, Lemma 3.18(2)]. Their interpretation method requires only weak monotonicity in this setting [1, Theorem 2.15].

The unary and mixed-base results therefore have different scopes: neither is a direct strengthening of the other. The new reversed result concerns the two eligible rules, not all eleven offsets. It does not exclude the forward boundary-rule TOP route, marked dependency-pair transformations of the mixed-base system, higher dimensions, or arctic interpretations.

The proof mechanisms also differ. The unary argument represents a configuration value by an $\mathbb{N}$ -rational sequence $u^T M^n v$ and contradicts its eventual growth properties [1, Theorems 3.5–3.8]. Our proof uses the mixed-base swap inequalities to force a common triangular orientation, then eliminates the affine gaps. The unrestricted reversed argument also compares growth along binary prefixes with bounds for ternary representations of the same integers. The relevant rule-removal framework and the distinction between extended and weak monotonicity are due to Endrullis, Waldmann, and Zantema [6, Theorems 2–3]. Their matrix criterion separates the coefficient comparisons from strictness of the first offset [6, Lemma 4]; Section 2 gives the fixed-decrement real version used here. The contributions are the two precise coefficient-independent obstructions for the mixed-base system and their Lean proofs.

2 The system and the interpretation format

We use the alphabet $\{a, b, c, d, e, f, g\}$ and the ASCII names from the authors’ implementation [2]. Table 1 relates these names to the symbols in their article. The rules are their full mixed-base

system [1, Section 3.2], with the leftmost symbol interpreted as the outermost map:

$$\begin{aligned}
 ad &\rightarrow d, & bd &\rightarrow gd, \\
 ae &\rightarrow ea, & af &\rightarrow eb, & ag &\rightarrow fa, \\
 be &\rightarrow fb, & bf &\rightarrow ga, & bg &\rightarrow gb, \\
 ce &\rightarrow cb, & cf &\rightarrow caa, & cg &\rightarrow cab.
 \end{aligned} \tag{1}$$

This paper and the ASCII implementation	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
Yolcu–Aaronson–Heule [1]	f	t	◁	▷	0	1	2

Table 1: Symbol correspondence. The letters a, b are binary digits, e, f, g are ternary digits, and c, d mark the boundaries.

The five symbols a, b, e, f, g are called digits; c, d are boundary symbols. Assign to each symbol s an affine map $F_s(x) = M_s x + v_s$ on $\mathbb{R}^2$. An interpretation is *admissible* when

$$M_s \in \mathbb{R}_{\geq 0}^{2 \times 2}, \quad v_s \in \mathbb{R}_{\geq 0}^2, \quad (M_s)_{00} \geq 1 \quad \text{for every } s. \tag{2}$$

For words, composition determines M_w, v_w ; in particular, $M_{uv} = M_u M_v$ and $v_{uv} = M_u v_v + v_u$. A rule $\ell \rightarrow r$ is *weak* if $M_\ell \geq M_r$ and $v_\ell \geq v_r$ entrywise. Its first-offset gap is $\text{gap}(\ell, r) = (v_\ell)_0 - (v_r)_0$. The strictness criterion considered here is $\text{gap}(\ell, r) > 0$.

Theorem 1. *For every admissible interpretation in two real coordinates, weak orientation of all eleven rules in (1) implies $\text{gap}(\ell, r) = 0$ for every rule. The same assertion holds for the eleven rules $\text{rev}(\ell) \rightarrow \text{rev}(r)$, where each word is reversed and the direction of every rule is retained.*

The two orientations are separate implications, not simultaneous hypotheses. There is no coefficient bound, integrality requirement, nonsingularity assumption, or prescribed boundary shape. The conclusion concerns the first offsets; it does not assert equality of the complete affine maps. Sections 3–6 give the matrix classifications and affine case analyses that prove this theorem. The reversed system admits a stronger result for its two dynamic rules.

Theorem 2 (Reversed TOP obstruction). *Let the seven symbol interpretations be nonnegative real affine maps on $\mathbb{R}^2$, with no diagonal lower bound. If all eleven word-reversed rules are weakly oriented coefficientwise, then*

$$v_{da} = v_d, \quad v_{db} = v_{dg}.$$

These are equalities of the entire offset vectors. In particular, neither $da \rightarrow d$ nor $db \rightarrow dg$ has a positive offset gap in either coordinate.

Section 7 proves this result. It does not claim that the other nine reversed gaps vanish without the diagonal bounds. For example, take a, b to be the identity, c, d the constant zero map, and e, f, g the same positive translation. All reversed rules are weak, but the three rules with inner symbol c have positive offset gaps.

2.1 Scope and soundness

Theorem 1 concerns the strictly monotone, full-context format in (2). Every symbol, including both boundaries, must satisfy $(M_s)_{00} \geq 1$. Theorem 2 instead concerns the two rules eligible for the reversed relative top-removal step in Yolcu–Aaronson–Heule’s Lemma 3.18(2) [1]. At the root, strict decrease need not be preserved by every surrounding symbol, so only weak monotonicity is required. The result excludes this first reversed TOP step using two nonnegative real affine coordinates and the strictness criterion below. It does not resolve the forward TOP problem, arctic interpretations, other strict orders, or higher dimensions.

Real coefficients require a fixed positive decrement. For $\delta > 0$, put

$$x \succ_{\delta} y \iff x_0 \geq y_0 + \delta \text{ and } x_1 \geq y_1, \quad x, y \in \mathbb{R}_{\geq 0}^2.$$

Along a decrease, $\lfloor x_0/\delta \rfloor$ strictly decreases in $\mathbb{N}$, so $\succ_{\delta}$ is well-founded. Nonnegative affine maps preserve coordinatewise weak order. The bound $(M_s)_{00} \geq 1$ also preserves $\succ_{\delta}$, since

$$(F_s(x) - F_s(y))_0 = (M_s)_{00}(x_0 - y_0) + (M_s)_{01}(x_1 - y_1) \geq \delta.$$

A weak rule with first-offset gap $\varepsilon > 0$ decreases by at least ε at every nonnegative input. For a finite nonempty set of strict rules, the least such gap supplies a common $\delta > 0$. These observations explain the real-coefficient certificate format; ordinary strict inequality on the reals alone would not suffice. Theorem 1 prevents even this first choice of a strict rule. The preservation, evaluation, and well-foundedness claims above are formalized separately in `FullTwoSoundness.lean`. For the root comparisons of Theorem 2, a positive uniform decrement at every nonnegative input would in particular hold at zero. The offset equalities rule this out without any assumption that strict decrease is preserved inside a context.

3 Reduction of the digit matrices

Write A, B, C, D, E, F, G for the seven matrices. Throughout this section, all seven matrices are entrywise nonnegative and have $(0, 0)$ entry at least one. We first treat the forward orientation. Its six digit comparisons are

$$AE \geq EA, \quad AF \geq EB, \quad AG \geq FA, \quad BE \geq FB, \quad BF \geq GA, \quad BG \geq GB. \quad (3)$$

The five boundary comparisons are

$$AD \geq D, \quad BD \geq GD, \quad CE \geq CB, \quad CF \geq CA^2, \quad CG \geq CAB. \quad (4)$$

All inequalities in this section are entrywise. In particular, they may be multiplied on either side by a nonnegative matrix. None of the arguments below uses an affine offset.

Lemma 3. *Suppose seven nonnegative real 2×2 matrices have $(0, 0)$ entries at least one and satisfy (3) and (4). Then the five digit matrices A, B, E, F, G are all upper triangular or all lower triangular in the original coordinates, and*

$$A_{00} = B_{00} = E_{00} = F_{00} = G_{00} = 1.$$

The same conclusion holds for the matrix comparisons of the reversed rules.

The distinction between triangularity in the original coordinates and triangularity after a change of basis matters: the order and the distinguished coordinate in (2) are fixed. We prove the lemma by excluding the case in which the sum of the digit matrices has both off-diagonal entries positive. The required algebraic classifications, including the singular cases, are given below.

3.1 From weak swaps to exact swaps

Put $P = A + B$ and $H = A + B + E + F + G$. The sum of the six differences in (3) is

$$K = PH - HP \geq 0.$$

Since $\text{tr } K = 0$, both diagonal entries of K vanish. Cyclicity of the trace also gives $\text{tr}(HK) = 0$. Consequently

$$H_{01}K_{10} + H_{10}K_{01} = 0.$$

If $H_{01}, H_{10} > 0$, this forces $K_{01} = K_{10} = 0$. Each of the six nonnegative differences that sum to K must then be zero. We call the resulting identities the *exact swaps*:

$$AE = EA, \quad AF = EB, \quad AG = FA, \quad BE = FB, \quad BF = GA, \quad BG = GB. \quad (5)$$

Even without the strict inequalities on H , the same trace argument shows that every diagonal entry of every swap difference is zero. We will use this weaker conclusion after proving triangularity.

We use the following elementary identities for real 2×2 matrices:

$$X^2 = (\operatorname{tr} X)X - (\det X)I, \quad (6)$$

$$RXR = \operatorname{tr}(RX)R + (\det R)X - (\det R)(\operatorname{tr} X)I, \quad (7)$$

$$XY + YX = (\operatorname{tr} X)Y + (\operatorname{tr} Y)X + (\operatorname{tr}(XY) - \operatorname{tr} X \operatorname{tr} Y)I. \quad (8)$$

The first is Cayley–Hamilton; the other two follow by expanding the four entries. Thus, if $\det R = 0$ and $\beta = \operatorname{tr} R \neq 0$, then $R^2 = \beta R$ and $RXR = \operatorname{tr}(RX)R$. These rank-one identities account for the singular branches of the proof.

3.2 The boundary comparison for a common rank-one action

The same scalar comparison will be used twice.

Lemma 4. *Assume the standing nonnegativity and first-diagonal hypotheses and (4). Let $R \geq 0$ satisfy $R_{00} > 0$ and, for real numbers α, β ,*

$$RA = RB = AR = \alpha R, \quad RG = FR = \beta R.$$

Then $\alpha = \beta = 1$.

Proof. The $(0,0)$ entry of $RA = \alpha R$ gives $\alpha \geq A_{00} \geq 1$. Also $(RD)_{00} > 0$ and $(CR)_{00} > 0$, because $C_{00}, D_{00} \geq 1$. Multiplying $BD \geq GD$ on the left by R gives $\alpha RD \geq \beta RD$, hence $\alpha \geq \beta$. Multiplying $CF \geq CA^2$ on the right by R gives $\beta CR \geq \alpha^2 CR$, hence $\beta \geq \alpha^2$. Therefore

$$1 \leq \alpha, \quad \alpha^2 \leq \beta \leq \alpha,$$

which implies the assertion. $\square$

We also need a specifically two-dimensional observation about stationary vectors. It supplies a right action from the left action arising in the singular-middle case.

Lemma 5. *Let $A, B \geq 0$ be 2×2 matrices with $A_{00}, B_{00} > 0$. Suppose w is a strictly positive row, $\alpha > 0$, $wA = wB = \alpha w$, and a column u satisfies*

$$ABu = BAu = \alpha^2 u.$$

Then $Au = Bu = \alpha u$.

Proof. Both matrices preserve the one-dimensional space $\ker w$. Their actions on that space are multiplication by

$$\kappa_A = \operatorname{tr} A - \alpha, \quad \kappa_B = \operatorname{tr} B - \alpha.$$

The positive left eigenrow and nonnegativity give $A_{ii}, B_{ii} \leq \alpha$; the positive $(0,0)$ entries give

$$-\alpha < \kappa_A, \kappa_B \leq \alpha.$$

Set $x = (A - \alpha I)u$ and $y = (B - \alpha I)u$. These vectors belong to $\ker w$, and the two product equations become

$$\alpha x + \kappa_A y = 0, \quad \alpha y + \kappa_B x = 0.$$

If $\kappa_A < \alpha$, the displayed bounds imply $\kappa_A \kappa_B < \alpha^2$, so $x = y = 0$. If $\kappa_A = \alpha$, both diagonal entries of A equal α ; $wA = \alpha w$ then makes both off-diagonal entries zero. Thus $A = \alpha I$, and $ABu = \alpha^2 u$ gives the same conclusion. $\square$

3.3 A singular middle digit

Here is the precise singular classification needed for the proof.

Lemma 6. *Let $A, B, E, F, G \geq 0$ satisfy the exact swaps and have strictly positive $(0, 0)$ entries. If $\det F = 0$, then, with $\beta = \operatorname{tr} F > 0$, there is $\alpha > 0$ such that*

$$AB = BA, \quad FA = FB = \alpha F, \quad FE = F^2 = FG = \beta F, \quad (9)$$

$$ABF = BAF = \alpha^2 F. \quad (10)$$

If in addition $H_{01}, H_{10} > 0$, then $F_{01}, F_{10} > 0$ and $AF = BF = \alpha F$.

Proof. The exact swaps give

$$ABF = FA^2, \quad BAF = FB^2.$$

Using $F^2 = \beta F$ in the first identity gives $FA^2F = \beta FA^2$. Cayley–Hamilton for A , whose trace is positive, reduces this equation to $FAF = \beta FA$. The rank-one sandwich identity therefore gives $FA = aF$, where $a = \operatorname{tr}(FA)/\beta$. Similarly $FB = bF$. Their $(0, 0)$ entries show $a, b > 0$.

Let $L = AB - BA$. Then $FL = 0$ and $\operatorname{tr} L = 0$. Equation (8) applied to F, L gives $LF = \beta L$. On the other hand, the two product identities give $LF = (a^2 - b^2)F$. Taking traces yields $a^2 = b^2$, hence $a = b = \alpha$, and then $L = 0$. Multiplying $BE = FB$ and $AG = FA$ on the left by F now gives $FE = FG = \beta F$. This proves (9)–(10).

These actions imply $FH = (2\alpha + 3\beta)F$. If $F_{01} = 0$, its $(0, 1)$ entry would give $F_{00}H_{01} = 0$. Thus the row $w = (F_{00}, F_{01})$ is strictly positive. Apply Lemma 5 to each column of F to obtain $AF = BF = \alpha F$.

To see that $F_{10} > 0$, suppose instead that $F_{10} = 0$. The determinant condition and $F_{00} > 0$ give $F_{11} = 0$. The $(1, 0)$ entries of $AG = FA$ and $BE = FB$, together with $G_{00}, E_{00} > 0$, then force $A_{10} = B_{10} = 0$. The same entries of $AF = EB$ and $BF = GA$ force $E_{10} = G_{10} = 0$. This contradicts $H_{10} > 0$. $\square$

Return to the standing first-diagonal bounds. Apply Lemma 4 with $R = F$ to obtain $\alpha = \beta = 1$. But $\operatorname{tr} F = 1$ and $F_{00} \geq 1$ imply $F_{11} = 0$; then

$$0 = \det F = -F_{01}F_{10} < 0.$$

This excludes a singular F whenever both off-diagonals of H are positive.

3.4 Singular binary digits with an invertible middle digit

Assume next that $\det F \neq 0$. Taking determinants in $AF = EB$ and $BE = FB$ gives $\det A = \det B$. If this common determinant is nonzero, $AG = FA$ and $BE = FB$ also give $\det E = \det F = \det G \neq 0$. The remaining singular branch is therefore exactly $\det A = \det B = 0$.

Lemma 7. *Let real 2×2 matrices satisfy the exact swaps, with*

$$\det A = \det B = 0, \quad \det F \neq 0, \quad \operatorname{tr} A, \operatorname{tr} B, \operatorname{tr} F > 0.$$

If $A \neq B$, there are $\alpha, \beta > 0$ such that

$$\begin{aligned} \operatorname{tr} A = \operatorname{tr} B = \alpha, \quad A^2 = AB = \alpha A, \quad BA = B^2 = \alpha B, \\ F = \beta I, \quad E = (\beta/\alpha)A, \quad G = (\beta/\alpha)B. \end{aligned} \quad (11)$$

If instead $A = B \geq 0$, $E \geq 0$, and $A_{00}, E_{00} > 0$, there is $\beta > 0$ such that, for $\alpha = \operatorname{tr} A > 0$,

$$A^2 = \alpha A, \quad AE = EA = AF = FA = AG = GA = \beta A. \quad (12)$$

Proof. Since F is invertible, $ABF = FA^2$ and $BAF = FB^2$ imply

$$\operatorname{tr}(AB) = (\operatorname{tr} A)^2 = (\operatorname{tr} B)^2.$$

The positive traces are equal; call their value α . For $N = B - A$, the two-dimensional identity

$$\det(B - A) = \det A + \det B + \operatorname{tr}(AB) - \operatorname{tr} A \operatorname{tr} B$$

gives $\operatorname{tr} N = \det N = \operatorname{tr}(AN) = 0$. If $A \neq B$, then $N \neq 0$ and $N^2 = 0$. Its image and kernel are the same line. The sandwich identity gives $NAN = 0$, so A preserves this line. Consequently

$$AN = aN, \quad NA = (\alpha - a)N$$

for some a . The second identity follows from (8); $A^2 = \alpha A$ makes $a = 0$ or $a = \alpha$.

The value $a = \alpha$ would give $AB = \alpha B$ and $BA = \alpha A$. The two intertwining identities with F would then yield $BF = FA$ and $AF = FB$, hence $FN + NF = 0$. Taking traces gives $\operatorname{tr}(FN) = 0$, and (8) reduces to $(\operatorname{tr} F)N = 0$, a contradiction. Thus $a = 0$, proving $AB = \alpha A$ and $BA = \alpha B$. The same intertwining identities now say that F commutes with both A and B .

A nonscalar 2×2 matrix has centralizer $\{xI + yA : x, y \in \mathbb{R}\}$. To see this, choose a column v for which v, Av are independent. A commuting matrix is determined by its value $xv + yAv$ on v , since its value on Av must then be $A(xv + yAv)$. Here A is nonscalar because it is singular with positive trace. Writing $F = xI + yA$, commutation with B gives $y\alpha(A - B) = 0$, so $F = \beta I$; its positive trace gives $\beta > 0$. Write $E = x'I + y'A$ using $AE = EA$. Then $AF = EB$ gives

$$\beta A = x'B + y'\alpha A.$$

The distinct matrices A, B with the same nonzero trace are linearly independent, so $x' = 0$ and $y' = \beta/\alpha$. The corresponding argument with $BG = GB$ and $BF = GA$ gives G .

For $A = B$, use $AE = EA$, $A^2 = \alpha A$, and $AEA = \operatorname{tr}(AE)A$. These give $AE = EA = \beta A$, where $\beta = \operatorname{tr}(AE)/\alpha > 0$; positivity follows from the $(0, 0)$ entry. The remaining exact swaps give all the other identities in (12). $\square$

We now exclude both alternatives under the standing boundary and first-diagonal hypotheses. If $A = B$, use $R = A$ in Lemma 4. It gives $\alpha = \beta = 1$. Equation (12) also gives

$$AH = HA = (2\alpha + 3\beta)A.$$

The $(0, 1)$ and $(1, 0)$ entries force $A_{01}, A_{10} > 0$ when $H_{01}, H_{10} > 0$. This contradicts $\operatorname{tr} A = 1$, $A_{00} \geq 1$, and $\det A = 0$.

If $A \neq B$, substitute (11) in $BD \geq GD$. Since $(BD)_{00} > 0$, it gives $\beta \leq \alpha$. Multiplying $CF \geq CA^2$ on the right by A and using $(CA)_{00} > 0$ gives $\beta \geq \alpha^2$. Also $\alpha = \operatorname{tr} A \geq 1$, so again $\alpha = \beta = 1$. The first-diagonal bounds now force

$$A_{00} = B_{00} = 1, \quad A_{11} = B_{11} = 0, \quad E = A, \quad F = I, \quad G = B.$$

The two determinant equations and the $(0, 0)$ entries of $AB = A$ and $BA = B$ give, respectively,

$$A_{01}A_{10} = B_{01}B_{10} = A_{01}B_{10} = B_{01}A_{10} = 0.$$

Hence $(A_{01} + B_{01})(A_{10} + B_{10}) = 0$, whereas both factors are positive if $H_{01}, H_{10} > 0$. This excludes the singular-binary branch.

3.5 The nonsingular classification and its resonance

It remains to consider invertible A, B, E, F, G .

Lemma 8. *Let real invertible 2×2 matrices satisfy the exact swaps, and suppose $\operatorname{tr} A, \operatorname{tr} B, \operatorname{tr} E > 0$. If $A = B$, then $E = F = G$ and $AE = EA$. If $A \neq B$, there are $\mu, t > 0$ such that*

$$\begin{aligned} \operatorname{tr} A = \operatorname{tr} B = 3\mu, \quad \det A = \det B = 2\mu^2, \\ A(B - A) = \mu(B - A), \quad (B - A)A = 2\mu(B - A), \\ E = t(2A - \mu I), \quad F = t(A + B - \mu I), \quad G = t(2B - \mu I). \end{aligned} \quad (13)$$

Proof. When $A = B$, cancellation of the invertible A in $AE = EA = AF$ and $AG = FA$ gives $E = F = G$.

Suppose $A \neq B$. Determinants give $\det A = \det B$. The exact swaps also give

$$(AB)E = EB^2, \quad (BA)G = GA^2.$$

Taking traces after cancellation of the invertible intertwiners, and using $\operatorname{tr}(X^2) = (\operatorname{tr} X)^2 - 2\det X$, gives

$$\operatorname{tr} A = \operatorname{tr} B, \quad \operatorname{tr}(AB) = \operatorname{tr}(A^2).$$

Thus $N = B - A \neq 0$ again satisfies $\operatorname{tr} N = \det N = \operatorname{tr}(AN) = 0$, so $N^2 = 0$ and

$$AN = aN, \quad NA = \kappa N, \quad a + \kappa = \operatorname{tr} A.$$

Invertibility of A makes a, κ nonzero. Moreover A cannot be scalar: if it were, the exact swaps and invertibility would successively give $G = F$, $E = F$, and $A = B$. Its centralizer therefore gives $E = xI + yA$.

Put $b = x + ya$, $e = b/a$, and $g = e\kappa/a$. The identities $AF = EB$ and $AG = FA$ give

$$F = E + eN, \quad G = E + gN.$$

Substitute these formulas into $BE = FB$ and $BG = GB$ and use $N^2 = 0$. Comparison of the coefficients of the nonzero matrix N gives

$$y(\kappa - a) = e\kappa, \quad (g - y)(a - \kappa) = 0, \quad ga = e\kappa. \quad (14)$$

If $a = \kappa$, the first equation makes $e = 0$, hence $EN = 0$, contradicting invertibility of E . Therefore $g = y$. Also $y \neq 0$, since otherwise the same equations again give $EN = 0$. Equations (14) now imply

$$\kappa = 2a, \quad e = y/2, \quad x = -ya/2.$$

This is the 2 : 1 resonance responsible for the nonscalar family. Set $\mu = a$ and $t = y/2$. The positive trace of A gives $\mu > 0$; $\operatorname{tr} E = 4t\mu > 0$ gives $t > 0$. Applying Cayley–Hamilton for A to N gives $\det A = 2\mu^2$. Substitution yields all of (13). $\square$

First exclude the repeated case $A = B$, $E = F = G = T$. Let c be the first row of C , d the first column of D , and $z = Ad - d \geq 0$. The boundary comparisons imply

$$cA^2d \leq cTd \leq cAd, \quad \text{so} \quad cAz \leq 0.$$

Nonnegativity, $c_0 > 0$, and $A_{00} \geq 1$ give $z_0 = 0$ and $A_{01}z_1 = 0$. Write

$$A = \begin{pmatrix} a & b \\ r & s \end{pmatrix}, \quad T = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

Since $d_0 > 0$, $z_0 = (a - 1)d_0 + bd_1 = 0$ gives $a = 1$ and $bd_1 = 0$. Then $bz_1 = 0$ gives $br = 0$. The first coordinate of $Td \leq Ad$, with $e \geq 1$, gives $fd_1 = 0$. If $H_{01} = 2b + 3f > 0$, we obtain $d_1 = 0$. The second coordinate gives $g \leq r$, and $H_{10} = 2r + 3g > 0$ makes $r > 0$. Hence $b = 0$ and $f > 0$. But the $(0, 0)$ entry of $AT = TA$ says $bg = fr$, a contradiction.

For the unequal case, put

$$U = A - \mu I, \quad V = B - \mu I, \quad S = U + V.$$

These matrices are nonnegative. Indeed, if the diagonal entries of A are p, q , then $p + q = 3\mu$ and

$$(p - \mu)(q - \mu) = A_{01}A_{10} \geq 0.$$

Their sum is $\mu > 0$, so both $p - \mu, q - \mu$ are nonnegative. The same argument applies to B . The classification gives

$$\operatorname{tr} U = \operatorname{tr} V = \mu, \quad U^2 = UV = \mu U, \quad VU = V^2 = \mu V. \quad (15)$$

In particular $SU = SV = \mu S$ and $S^2 = 2\mu S$. The off-diagonal entries of H are $(1 + 3t)$ times those of S . Thus $S_{01}, S_{10} > 0$, and the $(0, 0)$ entry of $S^2 = 2\mu S$ also gives $S_{00} > 0$.

Multiply $BD \geq GD$ on the left by S . Since $SB = 2\mu S$, $SG = 3t\mu S$, and $(SD)_{00} > 0$, we obtain $3t \leq 2$. Now put

$$u = (CU)_{00}, \quad v = (CV)_{00}.$$

They are nonnegative and $u + v = (CS)_{00} > 0$. Multiplying $CE \geq CB$ on the right by U , and $CG \geq CAB$ on the right by V , gives

$$u + v \leq 3tu, \quad 2\mu(u + v) \leq 3tv. \quad (16)$$

For example, the products used here are $BU = \mu S$, $EU = 3t\mu U$, $ABV = 2\mu^2 S$, and $GV = 3t\mu V$; these follow directly from (15). Together with $3t \leq 2$, (16) gives $v \leq u$ and $\mu(u + v) \leq v$, hence $\mu \leq 1/2$. On the other hand, $A_{00} = U_{00} + \mu \geq 1$ and $U_{00} + U_{11} = \mu$ give $2\mu \geq 1$. Therefore $\mu = 1/2$ and $U_{11} = V_{11} = 0$. The $(1, 1)$ entry of $S^2 = 2\mu S$ now reads $S_{10}S_{01} = 0$, again a contradiction.

3.6 The first diagonals in the triangular cases

We have excluded $H_{01}, H_{10} > 0$ in every determinant case. Nonnegativity therefore makes one off-diagonal entry zero in every digit matrix. This proves the common triangular orientation. The diagonal equalities obtained from the trace argument reduce, in either triangular orientation, to

$$A_{00}G_{00} = F_{00}A_{00}, \quad B_{00}E_{00} = F_{00}B_{00}, \quad A_{00}F_{00} = E_{00}B_{00}.$$

The positive first-diagonal bounds allow cancellation, so write

$$A_{00} = B_{00} = t \geq 1, \quad E_{00} = F_{00} = G_{00} = s \geq 1.$$

For completeness, the following entry arguments show why the boundary comparisons force $t = s = 1$, even when secondary entries vanish.

For upper triangular digits, write

$$A = \begin{pmatrix} t & \alpha \\ 0 & u \end{pmatrix}, \quad B = \begin{pmatrix} t & \beta \\ 0 & v \end{pmatrix}, \quad G = \begin{pmatrix} s & \gamma \\ 0 & y \end{pmatrix},$$

and let (c_0, c_1) be the first row of C , $(d_0, d_1)^\top$ the first column of D . Thus $c_0, d_0 \geq 1$ and all other displayed entries are nonnegative. The $(0, 0)$ entry of $CG \geq CAB$ gives $s \geq t^2$. If $s > t$,

the first coordinate of $BD \geq GD$ gives $d_1 > 0$ and $\beta > \gamma$. The second coordinates of $AD \geq D$ and $BD \geq GD$ give $u \geq 1$ and $v \geq y$. The $(0, 1)$ entry of $CG \geq CAB$ is

$$c_0(t\beta + \alpha v) + c_1 uv \leq c_0 \gamma + c_1 y.$$

Since $uv \geq y$, it implies $\beta \leq t\beta + \alpha v \leq \gamma$, a contradiction. Thus $t^2 \leq s \leq t$, proving $t = s = 1$.

For lower triangular digits, write

$$A = \begin{pmatrix} t & 0 \\ \alpha & u \end{pmatrix}, \quad B = \begin{pmatrix} t & 0 \\ \beta & v \end{pmatrix}, \quad E = \begin{pmatrix} s & 0 \\ \varepsilon & w \end{pmatrix},$$

$$F = \begin{pmatrix} s & 0 \\ \phi & x \end{pmatrix}, \quad G = \begin{pmatrix} s & 0 \\ \gamma & y \end{pmatrix}.$$

Now $BD \geq GD$ gives $s \leq t$. Suppose $t > 1$, so $t^2 > s$. The $(0, 0)$ entries of $CF \geq CA^2$, $CG \geq CAB$, and $CE \geq CB$ give

$$c_1 > 0, \quad \phi > (t + u)\alpha \geq \alpha, \quad \gamma > t\alpha + u\beta, \quad \beta \leq \varepsilon.$$

Their $(0, 1)$ entries give $v \leq w$, $u^2 \leq x$, and $uv \leq y$. If $u = 0$, the $(1, 0)$ swap comparison $AG \geq FA$ gives $t\phi + x\alpha \leq s\alpha$, contradicting $\phi > \alpha$ and $t \geq s$.

If $u > 0$, then $x > 0$. The diagonal swap equalities $uy = xu$, $vx = yu$, and $ux = wv$ give $y = w = x$ and $v = u$; hence $x \geq u$. The $(1, 0)$ entry of $BD \geq GD$ gives $\gamma \leq \beta$. Together with $\gamma > t\alpha + u\beta$, this forces $u < 1$. The $(1, 0)$ entry of $AE \geq EA$ now yields

$$(t - u)\varepsilon \leq (s - x)\alpha \leq (t - u)\alpha,$$

so $\varepsilon \leq \alpha$ and therefore $\gamma \leq \beta \leq \varepsilon \leq \alpha$. Finally $AG \geq FA$ gives

$$t\phi + x\alpha \leq s\alpha + u\gamma \leq (s + u)\alpha,$$

and $x \geq u$, $s \leq t$ imply $\phi \leq \alpha$, a contradiction. Thus $t \leq 1$, completing the lower triangular case.

Completion of the proof of Lemma 3. If both off-diagonals of H were positive, the trace argument would give the exact swaps. The singular-middle, singular-binary, repeated, and unequal nonsingular arguments above exhaust the determinant and equality cases and exclude that possibility. The final subsection then gives the common triangular orientation and the five unit first diagonals.

For reversed rules, transpose every symbol matrix. Since $(XY)^\top = Y^\top X^\top$, the reversed matrix comparisons become exactly (3) and (4), with the same symbol names. Transposition preserves nonnegativity and each $(0, 0)$ entry, and exchanges upper and lower triangularity. The forward conclusion therefore proves the reversed conclusion as well. $\square$

4 The lower triangular affine case

After Lemma 3, lower triangular digit maps have the form

$$F_s(x, y) = (x + h_s, q_s x + z_s y + k_s), \quad h_s, q_s, z_s, k_s \geq 0. \quad (17)$$

Their first offsets add under composition. The six weak swap gaps sum to zero, so all six vanish and

$$h_a = h_b = r, \quad h_e = h_f = h_g = u, \quad r, u \geq 0. \quad (18)$$

4.1 Scalar classifications used in the lower case

The scalar statements below also handle zero slopes. They are stated with the inequalities actually used, so their application does not require a scalar version of the diagonal lower bound.

Lemma 9 (Forward scalar classification). *Let $a, b, e, f, g, \alpha, \beta, \varepsilon, \varphi, \gamma, \delta$ be nonnegative real numbers. Suppose*

$$b \leq e, \quad a^2 \leq f, \quad af \geq eb, \quad ag \geq fa, \quad be \geq fb, \quad bf \geq ga, \quad (19)$$

$$\begin{aligned} b\delta + \beta &\geq g\delta + \gamma, & a\varepsilon + \alpha &\geq e\alpha + \varepsilon, \\ a\varphi + \alpha &\geq e\beta + \varepsilon, & a\gamma + \alpha &\geq f\alpha + \varphi, \end{aligned} \quad (20)$$

$$\begin{aligned} b\varepsilon + \beta &\geq f\beta + \varphi, & b\varphi + \beta &\geq g\alpha + \gamma, \\ \varepsilon &\geq \beta, & \varphi &\geq (a+1)\alpha, & \gamma &\geq a\beta + \alpha. \end{aligned} \quad (21)$$

Then either all five offsets $\alpha, \beta, \varepsilon, \varphi, \gamma$ are zero, or all five slopes a, b, e, f, g are zero and all five offsets are equal and positive. In particular,

$$\varepsilon = \beta, \quad \varphi = (a+1)\alpha, \quad \gamma = a\beta + \alpha.$$

Proof. The four nonnegative swap differences in (19) sum to zero. Hence $af = eb$, $ag = fa$, $be = fb$, and $bf = ga$.

If $a = 0$, then $eb = 0$ and $e \geq b$ give $b = 0$. The ag offset comparison and $\varphi \geq \alpha$ give $\varphi = \alpha$. The be comparison then gives $\beta \geq \alpha$. On the other hand the ae comparison gives $\alpha \geq e\alpha + \varepsilon \geq \varepsilon \geq \beta$. Thus $\alpha = \beta = \varepsilon = \varphi$. The bf comparison and $\gamma \geq \alpha$ also give $\gamma = \alpha$. If this common value is positive, the ae, ag, bf comparisons force $e = f = g = 0$. This proves the assertion in this branch.

If $a > 0$, the inequality $f \geq a^2$ gives $f > 0$; $af = eb$ then gives $b > 0$. The swap equalities force

$$a = b =: t > 0, \quad e = f = g =: s, \quad s \geq t, \quad s \geq t^2.$$

The first inequality in (20) implies $\gamma \leq \beta$, since $(s-t)\delta \geq 0$. If $t \geq 1$, the chain

$$\beta \geq \gamma \geq t\beta + \alpha \geq \beta + \alpha$$

gives $\alpha = 0$, $\gamma = \beta = t\beta$. The ag comparison gives $\varphi \leq \beta$, while af gives $s\beta + \varepsilon \leq t\varphi \leq t\beta = \beta$. Because $s \geq t \geq 1$, this forces $\varepsilon = 0$, hence $\beta = \gamma = \varphi = 0$.

If $t < 1$, the ae comparison gives $(1-t)\varepsilon \leq (1-s)\alpha \leq (1-t)\alpha$. Consequently $\beta \leq \varepsilon \leq \alpha \leq \gamma \leq \beta$. All these quantities are equal, and $\gamma \geq t\beta + \alpha$ with $t > 0$ forces their common value to be zero. The ag comparison finally gives $\varphi = 0$. $\square$

For application to maps $y \mapsto z_s y + q_s$, the five slopes above are z_a, z_b, z_e, z_f, z_g and the offsets are q_a, q_b, q_e, q_f, q_g . The dynamic comparison is evaluated at the nonnegative inner-boundary value δ . The same statement applies with k_s in place of q_s .

Lemma 10 (Reversed scalar cancellation). *Let $a \geq 1$ and let $b, f, g, \alpha, \beta, \varphi, \gamma, \xi$ be nonnegative. Suppose*

$$\begin{aligned} g &\leq b, & \gamma &\leq \beta, & af &\leq ag, \\ b(a\xi + \alpha) + \beta &\leq g\xi + \gamma, \\ a\varphi + \alpha &\leq g\alpha + \gamma, & a\gamma + \alpha &\leq f\beta + \varphi. \end{aligned}$$

Then $\alpha = 0$ and $\beta = \gamma$.

Proof. The first offset inequality says

$$(ba - g)\xi + b\alpha + (\beta - \gamma) \leq 0.$$

All three terms are nonnegative, so $b\alpha = 0$ and $\beta = \gamma$. If $\alpha > 0$, then $b = 0$; $g \leq b$ and $af \leq ag$ give $g = f = 0$. The remaining inequalities become $a\varphi + \alpha \leq \gamma$ and $a\gamma + \alpha \leq \varphi$. Their sum contradicts $a \geq 1$. $\square$

Lemma 11 (Exceptional reversed scalar shape). *Let $a \geq 1$, and let the other variables below be nonnegative. Assume*

$$g \leq b, \quad af = eb, \quad ag = fa, \quad be = fb,$$

and

$$b\varepsilon + \beta \leq \varphi, \quad a\varphi \leq \beta, \quad a\varepsilon \leq \varepsilon, \quad b\varphi + \beta \leq e\beta + \varepsilon.$$

Then either $\beta = \varepsilon = \varphi = 0$, or

$$a = 1, \quad b = f = g = 0, \quad \varphi = \beta.$$

Proof. The first two offset inequalities give $\varphi = \beta$ and $b\varepsilon = 0$. If $a > 1$, $a\varphi \leq \beta = \varphi$ and $a\varepsilon \leq \varepsilon$ give the zero case. If $a = 1$ and $b > 0$, then $\varepsilon = 0$ and $be = fb$ gives $e = f = g \leq b$. The last inequality then yields $b\beta + \beta \leq e\beta \leq b\beta$, so $\beta = \varphi = 0$. If $a = 1, b = 0$, the equalities $af = eb$ and $ag = fa$ give $f = g = 0$, which is the exceptional case. $\square$

4.2 Application to the two-coordinate maps

Proposition 12. *For lower digit maps (17), every first-offset gap vanishes in either orientation, with arbitrary admissible boundary maps.*

Forward orientation. The dynamic rule $bd \rightarrow gd$ gives $r \geq u$. Write the first row of C as (c_0, c_1) , with $c_0 \geq 1$. If $c_1 = 0$, the cf boundary gives $u \geq 2r$, hence $r = u = 0$, and every boundary gap is zero.

Suppose $c_1 > 0$. The lower rows of the digit matrix products define scalar affine maps $y \mapsto z_sy + q_s$. The matrix inequalities induce all the forward scalar constraints, with outer boundary the identity and inner boundary the constant D_{10}/D_{00} . Lemma 9 gives precisely two possibilities: every $q_s = 0$, or every $q_s = v > 0$ and every $z_s = 0$.

If every $q_s = 0$, apply Lemma 9 to $y \mapsto z_sy + k_s$. Put

$$\Delta_e = k_e - k_b, \quad \Delta_f = k_f - (1 + z_a)k_a, \quad \Delta_g = k_g - k_a - z_ak_b.$$

The original boundary offset inequalities give $c_1\Delta_e \geq c_0(r - u)$ and $c_1\Delta_f, c_1\Delta_g \geq c_0(2r - u)$. Since $r \geq u$, these supply the remaining scalar weak constraints. Lemma 9 makes all three Δ 's zero. Hence $u \geq 2r$, so $r = u = 0$ and all eleven first-offset gaps vanish.

In the other case, $q_s = v > 0$ and $z_s = 0$. The second-offset ag inequality and first-offset cf inequality yield

$$k_a - k_f \geq v(r - u), \quad k_f - k_a \geq vr + (c_0/c_1)(2r - u).$$

Their sum forces $r = u = 0$. The remaining weak inequalities make all five k_s equal, so the three boundary gaps vanish as well. $\square$

Reversed orientation. The first-offset swap equalities still give (18); now the inner boundary $fc \rightarrow aac$ gives $u \geq 2r$. If $D_{01} = 0$, the outer dynamic comparison gives $r \geq u$, and the result follows.

For $D_{01} > 0$, the two dynamic matrix comparisons give $z_a \geq 1$, $z_g \leq z_b$, and $q_g \leq q_b$. Write $c_0 = C_{00} > 0$, $c_1 = C_{10} \geq 0$. The gc matrix comparison reads

$$z_b(z_a c_1 + q_a c_0) + q_b c_0 \leq z_g c_1 + q_g c_0.$$

Together with the ga, fb matrix comparisons, this is Lemma 10 with

$$(\alpha, \beta, \varphi, \gamma, \xi) = (q_a c_0, q_b c_0, q_f c_0, q_g c_0, c_1).$$

It gives $q_a = 0$, $q_b = q_g$. The remaining ea, fa, ga, eb comparisons now supply Lemma 11. Thus either all $q_s = 0$, or

$$z_a = 1, \quad z_b = z_f = z_g = 0, \quad q_a = 0, \quad q_b = q_f = q_g = \beta \geq 0.$$

No claim about z_e or q_e is needed in the exceptional case.

In the zero case, the db offset comparison and $u \geq 2r$ give $k_g \leq k_b$. Apply Lemma 10 again, now to $(\alpha, \beta, \varphi, \gamma, \xi) = (k_a, k_b, k_f, k_g, (v_c)_1)$. It gives $k_a = 0$ and $k_b = k_g$, so db forces $r \geq u$. Hence $r = u = 0$.

In the exceptional case, the gb comparison gives

$$k_g - k_b \geq q_b(u - r) \geq 0.$$

Together with $D_{00}u + D_{01}k_g \leq D_{00}r + D_{01}k_b$, this forces $r = u = 0$ and $k_b = k_g$. The ga, fb offset comparisons reduce to $k_f + k_a \leq k_g$ and $k_g + k_a \leq k_f$; therefore $k_a = 0$. The dynamic gaps are now zero. The six swap and three inner-boundary first-offset gaps depend only on r, u , so they vanish as well. $\square$

5 Upper normalization and aggregate cancellation

Upper digit maps with unit first diagonals have the form

$$F_s(x, y) = (x + q_s y + h_s, z_s y + k_s), \quad q_s, z_s, h_s, k_s \geq 0. \quad (22)$$

In the forward orientation, replace the boundary maps by

$$\hat{C}(x, y) = (x + \kappa y, 0), \quad \hat{D}(x, y) = (x, \tau), \quad \kappa = C_{01}/C_{00}, \quad \tau = (v_d)_1. \quad (23)$$

The outer boundary matrix and offset comparisons in coordinate zero are divided by $C_{00} > 0$. The dynamic offset differences are unchanged, and their normalized matrix differences are zero. Thus all weak rules are preserved, and zero first-offset gaps in the normalized model imply zero gaps in the original model. This leaves 22 nonnegative parameters.

For a digit word, composition in (22) is described exactly by

$$\begin{aligned} q_{uv} &= q_v + q_u z_v, & z_{uv} &= z_u z_v, \\ h_{uv} &= h_u + h_v + q_u k_v, & k_{uv} &= k_u + z_u k_v. \end{aligned} \quad (24)$$

In particular, the products $q_u k_v$ remain in every offset comparison.

Lemma 13. *Put*

$$\begin{aligned} p &= z_a + z_b - 2, & Q &= q_a + q_b, & R &= k_a + k_b, \\ r &= z_e + z_f + z_g - 3, & P &= q_e + q_f + q_g, & K &= k_e + k_f + k_g. \end{aligned}$$

The sums of the six swap gaps in the q, k, h coordinates are

$$G_q = rQ - pP \geq 0, \quad G_k = pK - rR \geq 0, \quad G_h = QK - PR \geq 0. \quad (25)$$

If $p < 0$ or $r < 0$, and $Q + P, R + K > 0$, all six affine swaps are exact.

Proof. Expanding (24) gives (25) and

$$RG_q + QG_k = pG_h, \quad KG_q + PG_k = rG_h.$$

A negative coefficient on the right forces $G_h = 0$. Adding the resulting equalities gives $(R + K)G_q + (Q + P)G_k = 0$, hence $G_q = G_k = 0$. Every individual gap in these sums is nonnegative. The secondary-slope swap gaps also sum to zero, completing the assertion. $\square$

Lemma 14 (Degenerate off-diagonal or offset data). *For a normalized upper model satisfying all eleven weak rules, either of the conditions*

$$q_a = q_b = q_e = q_f = q_g = 0 \quad \text{or} \quad k_a = k_b = k_e = k_f = k_g = 0$$

implies that all eleven first-offset gaps vanish. There is no additional condition on the secondary slopes z_s .

Proof. If all $q_s = 0$, the first offsets add, so the four nontrivial swap inequalities give $h_a = h_b = r$ and $h_e = h_f = h_g = u$. The dynamic comparison gives $r \geq u$. For $\kappa = 0$, the cf comparison gives $u \geq 2r$, hence $r = u = 0$. For $\kappa > 0$, the ce, cf matrix comparisons give $z_e \geq z_b$, $z_f \geq z_a^2$. The boundary offset comparisons give

$$\kappa(k_e - k_b) \geq r - u, \quad \kappa(k_f - (1 + z_a)k_a) \geq 2r - u,$$

and the analogous inequality for $k_g - k_a - z_a k_b$. These supply the boundary hypotheses of Lemma 9 for the five maps $y \mapsto z_s y + k_s$, with inner value $\delta = \tau$. Its three zero-gap conclusions again give $u \geq 2r$. Thus $r = u = 0$, and the scalar equalities make the three outer gaps zero.

If all $k_s = 0$, the same swap argument gives the two values r, u . Now cf gives $u \geq 2r$. When $\tau = 0$, the dynamic comparison gives $r \geq u$. When $\tau > 0$, its matrix and offset components imply

$$z_a \geq 1, \quad z_g \leq z_b, \quad q_g \leq q_b.$$

The q -composition rule in (24) reverses the order of scalar affine composition. Thus the relevant q -inequalities are exactly those of Lemma 10, with

$$(a, b, f, g, \alpha, \beta, \varphi, \gamma, \xi) = (z_a, z_b, z_f, z_g, q_a, q_b, q_f, q_g, \kappa).$$

For example the cg matrix comparison is

$$z_b(z_a \kappa + q_a) + q_b \leq z_g \kappa + q_g,$$

and the ag, bf comparisons give its two remaining offset inequalities. The lemma gives $q_a = 0$ and $q_b = q_g$. Consequently bd gives $r \geq u$, and again $r = u = 0$. Substitution in (24) now makes all eleven gaps zero. $\square$

The remaining case has positive total q and positive total k . The next lemmas identify what makes the common-form branches impossible.

Lemma 15 (Projection of exact swaps). *Suppose a normalized upper model satisfies all weak rules and its six affine swaps are exact. If $\lambda \geq 0$ and*

$$q_s = \lambda(1 - z_s) \quad (s = a, b, e, f, g),$$

put $H_s = h_s + \lambda k_s$. Then

$$H_a = H_b =: L, \quad H_e = H_f = H_g =: N, \quad 0 \leq N \leq L.$$

Proof. The nonnegative row $\ell(x, y) = x + \lambda y$ satisfies

$$\ell(F_s(x, y)) = \ell(x, y) + H_s.$$

Apply ℓ to the exact ag, be, af swaps, in that order. Their constant terms give $H_g = H_f$, $H_e = H_f$, and $H_a = H_b$. Applying ℓ to the weak dynamic comparison $F_b \widehat{D} \geq F_g \widehat{D}$ gives $L \geq N$. $\square$

Lemma 16 (Common fixed height). *Under the hypotheses of Lemma 15, assume $\lambda > 0$ and $k_s = \eta(1 - z_s)$ for every digit, where $\eta \geq 0$. Then all five k_s are zero.*

Proof. Every digit fixes the height $y = \eta$. At this height,

$$F_s(x, \eta) = (x + H_s, \eta),$$

because $q_s \eta = \lambda k_s$. Evaluate the weak cf boundary comparison at $(0, \eta)$. The common term $\kappa \eta$ cancels, leaving $N \geq 2L$. Together with $0 \leq N \leq L$, this gives $L = N = 0$. Each $H_s = h_s + \lambda k_s$ is a sum of nonnegative terms and $\lambda > 0$, so every $k_s = 0$. $\square$

Lemma 17 (Matching boundary projection). *Under the hypotheses of Lemma 15, if $\lambda > 0$ and $\kappa = \lambda$, then all five k_s are zero. No common-form hypothesis on the k_s is needed.*

Proof. The first component of $\widehat{C}$ is now ℓ . The cf offset comparison therefore gives $N \geq 2L$ directly. The last part of the preceding proof applies. $\square$

6 Completion of the upper case and word reversal

We continue with the normalized maps (22)–(23). All their parameters are nonnegative. The first diagonal of every digit matrix is one; no positivity is assumed for its secondary slope z_s . The next lemma explains the two exceptional slope pairs that occur in the proof.

Lemma 18 (Triangular resonance). *Let $\alpha, \beta, \rho > 0$, and let σ, a, b, e, f, g be real numbers. Consider two upper triangular matrices with common diagonal (α, ρ) and upper entries a, b , and three with common diagonal (β, σ) and upper entries e, f, g . Their six forward swap equalities on the upper entries are*

$$\begin{aligned} \alpha e + a\sigma &= \beta a + e\rho, & \alpha f + a\sigma &= \beta b + e\rho, \\ \alpha g + a\sigma &= \beta a + f\rho, & \alpha e + b\sigma &= \beta b + f\rho, \\ \alpha f + b\sigma &= \beta a + g\rho, & \alpha g + b\sigma &= \beta b + g\rho. \end{aligned} \tag{26}$$

These equations hold if and only if at least one of the following descriptions holds:

$$a = b, \quad e = f = g, \quad (\alpha - \rho)e = (\beta - \sigma)a; \tag{27}$$

$$\rho = 2\alpha, \quad \sigma = 3\beta, \quad \alpha e = 2\beta a, \quad \alpha f = \beta(a + b), \quad \alpha g = 2\beta b. \tag{28}$$

The two descriptions may overlap. In particular, if $a \neq b$, the exceptional diagonal ratios in (28) are necessary. No sign assumption on σ or on the five upper entries is needed for this algebraic statement.

Proof. Put $d = b - a$, $u = f - e$, and $v = g - e$. Subtracting the first equation in (26) from the second, fourth, third, and sixth, respectively, gives

$$\alpha u = \beta d, \quad (\sigma - \beta)d = \rho u, \quad \alpha v = \rho u, \quad (\alpha - \rho)v = (\beta - \sigma)d. \tag{29}$$

If $d = 0$, the first and third identities, together with $\alpha > 0$, give $u = v = 0$. The first equation of (26) is then the relation in (27).

If $d \neq 0$, positivity of α, β, ρ makes both u and v nonzero. The second and fourth identities give $(\alpha - \rho)v = -\rho u$; using the third yields $(2\alpha - \rho)v = 0$. Thus $\rho = 2\alpha$. The first two identities now give $(\sigma - 3\beta)d = 0$, so $\sigma = 3\beta$. Substitution into the first three original equations gives $\alpha e = 2\beta a$, $\alpha f = \beta(a + b)$, and $\alpha g = 2\beta b$. Finally, substitution of either (27) or (28) verifies all six equations, including the fifth equation, which was not needed to derive the alternatives. $\square$

For the q entries in (22), equal binary secondary slopes $t > 0$ and equal ternary secondary slopes $s \geq 0$ give $(\alpha, \beta, \rho, \sigma) = (1, 1, t, s)$ in the lemma. Their exceptional pair is therefore $(t, s) = (2, 3)$. For the k entries, use the homogeneous matrices of the scalar maps $y \mapsto z_s y + k_s$. Here the four diagonal parameters are $(t, s, 1, 1)$, so the exceptional pair is $(t, s) = (1/2, 1/3)$. The latter use requires $t, s > 0$; zero slopes will be handled separately rather than divided out.

Proposition 19. *Every normalized upper model satisfying all eleven weak forward rules has zero first-offset gap in every rule.*

Proof. If all q_s or all k_s vanish, Lemma 14 applies. Hence it suffices to rule out a model with

$$Q + P > 0, \quad R + K > 0, \quad (30)$$

using the notation of Lemma 13. We divide the proof according to the secondary slopes.

1. *Both binary slopes are positive.* The secondary-slope swap equalities are

$$z_a z_f = z_e z_b, \quad z_a z_g = z_f z_a, \quad z_b z_e = z_f z_b, \quad z_b z_f = z_g z_a. \quad (31)$$

Thus $z_e = z_f = z_g = s \geq 0$. If $s > 0$, the first equality also gives $z_a = z_b = t > 0$.

First suppose $s > 0$ and $t < 1$ or $s < 1$. Then $p = 2(t - 1) < 0$ or $r = 3(s - 1) < 0$, so Lemma 13 makes all six affine swaps exact. Apply Lemma 18 to their q entries. The exceptional pair $(2, 3)$ is incompatible with the present case. Consequently

$$q_a = q_b = x, \quad q_e = q_f = q_g = y, \quad (1 - t)y = (1 - s)x.$$

Here $x, y \geq 0$ and $x + y > 0$. These conditions give a parameter $\lambda > 0$ with

$$q_s = \lambda(1 - z_s) \quad (s \in \{a, b, e, f, g\}). \quad (32)$$

Indeed, if $t < 1$, then $x > 0$ and one can take $\lambda = x/(1 - t)$; otherwise $s < 1$, $y > 0$, and one takes $\lambda = y/(1 - s)$.

The resonance lemma applied to the k entries now gives either the same common form $k_s = \eta(1 - z_s)$, with $\eta > 0$ by (30), or $(t, s) = (1/2, 1/3)$. The common-form branch is excluded by Lemma 16. In the exceptional branch, the matrix gaps of $ce \rightarrow cb$ and $cf \rightarrow caa$, after substituting (32), are

$$\frac{\lambda - \kappa}{6} \geq 0, \quad \frac{\kappa - \lambda}{12} \geq 0.$$

Thus $\kappa = \lambda$, and Lemma 17 forces every k_s to vanish. This also contradicts (30). The exceptional resonance therefore does not leave an untreated family of offsets.

It remains, for $s > 0$, to consider $s, t \geq 1$. If $s > t$, the secondary-offset comparisons for $bd \rightarrow gd$ and $bg \rightarrow gb$ give

$$k_b \geq k_g + (s - t)\tau \geq k_g, \quad (t - 1)k_g \geq (s - 1)k_b.$$

Since $s - 1 > t - 1 \geq 0$, these force $k_b = k_g = \tau = 0$. The $ag \rightarrow fa$ comparison becomes $(1 - s)k_a \geq k_f$, so $k_a = k_f = 0$; $af \rightarrow eb$ then gives $k_e = 0$.

If $1 \leq s \leq t$ and $t > 1$, the three boundary matrix comparisons give

$$q_e \geq q_b, \quad q_f \geq (1 + t)q_a, \quad q_g \geq q_b + tq_a.$$

The discarded terms are nonnegative multiples of $t - s$ or $t^2 - s$. In particular $P \geq 2Q$. On the other hand, the aggregate q inequality is

$$2(t - 1)P \leq 3(s - 1)Q \leq 3(t - 1)Q.$$

Dividing by $t - 1 > 0$ gives $P \leq 3Q/2$. Thus $Q = P = 0$, contrary to (30).

For the remaining unit case $s = t = 1$, the linear swap inequalities in the q and k coordinates force

$$q_a = q_b = q, \quad q_e = q_f = q_g = p_0, \quad k_a = k_b = k, \quad k_e = k_f = k_g = \ell.$$

The boundary matrix comparison $cf \rightarrow caa$, the secondary-offset comparison $bd \rightarrow gd$, and the first-offset comparison $ae \rightarrow ea$ then yield, respectively,

$$p_0 \geq 2q, \quad k \geq \ell, \quad p_0k \leq q\ell.$$

Hence $2qk \leq p_0k \leq q\ell \leq qk$, and therefore $qk = p_0k = 0$. Positive total q implies $p_0 > 0$, whereas positive total k implies $k > 0$. This contradiction completes the unit case. Without the positive-total assumption, these same inequalities place the model in one of the two degenerate cases of Lemma 14.

Finally, suppose $s = 0$. The binary slopes z_a, z_b may now differ. Since $r = -3 < 0$, all swaps are exact. The af, be and ag, bf pairs give

$$(1 + z_b)(q_f - q_e) = 0, \quad (1 + z_a)(q_g - q_f) = 0.$$

Thus $q_e = q_f = q_g = \lambda$, and the ae, bg equalities give $q_a = \lambda(1 - z_a)$ and $q_b = \lambda(1 - z_b)$. Positive total q makes $\lambda > 0$. The corresponding k equalities give

$$k_e = k_f = k_g = \eta, \quad k_a = \eta(1 - z_a), \quad k_b = \eta(1 - z_b).$$

For example, subtracting the ae and af equalities gives $z_a(k_f - k_e) = 0$, and then ag gives $k_g = k_e$. Here $z_a > 0$ is available. Positive total k makes $\eta > 0$, so Lemma 16 again applies.

2. *At least one binary slope is zero.* Consider first $z_a = 0$ and $z_b > 1$. Equations (31) give $z_e = z_f = 0$. The af and be matrix comparisons become

$$q_f \geq q_b + z_b q_e, \quad q_e \geq q_b + z_b q_f.$$

Their sum gives $(1 - z_b)(q_e + q_f) \geq 2q_b$; nonnegativity and $z_b > 1$ force $q_b = q_e = q_f = 0$. The ae and bg comparisons then force $q_a = q_g = 0$. For $z_b = 0$ and $z_a > 1$, the slope equalities instead give $z_f = z_g = 0$. Now ag and bf yield

$$q_g \geq q_a + z_a q_f, \quad q_f \geq q_a + z_a q_g,$$

forcing $q_a = q_f = q_g = 0$; af and ae then give $q_b = q_e = 0$. Both cases contradict positive total q .

In the remaining zero-slope cases $z_a + z_b \leq 1$, so $p = z_a + z_b - 2 < 0$ and the swaps are exact. If exactly one binary slope is zero, their solutions have the common forms in the following table:

Binary slopes	Forced ternary zeros	Parameters in the common forms
$z_a = 0 < z_b$	$z_e = z_f = 0$	$\lambda = q_a, \eta = k_a$
$z_b = 0 < z_a$	$z_f = z_g = 0$	$\lambda = q_b, \eta = k_b$

For clarity, the first row means

$$q_e = q_f = q_a, \quad q_b = q_a(1 - z_b), \quad q_g = q_a(1 - z_g),$$

and the same identities with q replaced by k . The equality $q_e = q_a$ follows from ae ; the af, be pair gives $(1 + z_b)(q_f - q_a) = 0$, and ag gives the formula for q_g . The exact secondary-offset equations give the displayed k identities. For the second row one obtains

$$q_f = q_g = q_b, \quad q_a = q_b(1 - z_a), \quad q_e = q_b(1 - z_e),$$

again with the same k identities. Here af gives $q_f = q_b$, the ag, bf pair gives $(1 + z_a)(q_g - q_b) = 0$, and be determines q_e . Thus these conclusions use the actual forward equations in each case; no exchange of symbol labels is needed. Both parameters are positive under (30), and Lemma 16 excludes both rows of the table.

It remains to set $z_a = z_b = 0$. The exact af, bf matrix equations give $(1 + z_f)(q_a - q_b) = 0$, so

$$q_a = q_b = q > 0, \quad q_s = q(1 - z_s) \quad (s = e, f, g). \quad (33)$$

The six secondary-offset equations imply

$$z_e(k_a - k_b) = 0, \quad (1 - z_f)(k_a - k_b) = 0, \quad z_g(k_a - k_b) = 0. \quad (34)$$

If $k_a = k_b$, they give the common form with $\eta = k_a$, already excluded. Otherwise (34) forces exactly

$$(z_a, z_b, z_e, z_f, z_g) = (0, 0, 0, 1, 0), \quad k_e = k_a, \quad k_f = 0, \quad k_g = k_b.$$

Equation (33) gives $q_a = q_b = q_e = q_g = q$ and $q_f = 0$. The matrix comparison $cf \rightarrow caa$ now says $\kappa \geq q$; its offset comparison says

$$h_f \geq 2h_a + (q + \kappa)k_a \geq 2(h_a + qk_a).$$

Apply Lemma 15 with $\lambda = q$. The common binary projected offset is $L = h_a + qk_a$, and the common ternary projected offset is $N = h_f + qk_f = h_f$. The dynamic comparison gives $L \geq N$, while the last display gives $N \geq 2L$. Their nonnegativity forces $L = N = 0$. Since $q > 0$, every projected offset $h_s + qk_s = 0$ forces $k_s = 0$, contradicting positive total k .

The cases exhaust all nonnegative secondary slopes. Thus every weak normalized upper model lies in a case covered by Lemma 14, which supplies all eleven first-offset equalities. $\square$

6.1 Reversal after boundary normalization

For a reversed upper model, normalize the boundaries as

$$\hat{D}(x, y) = (x + \lambda y, 0), \quad \hat{C}(x, y) = (x, \sigma), \quad \lambda = \frac{D_{01}}{D_{00}}, \quad \sigma = (v_c)_1. \quad (35)$$

The denominator is positive because the original boundary is admissible. Each dynamic rule has common outer symbol d , so its first-row matrix comparison and first-offset gap are divided by D_{00} . Each of the other three boundary rules has common inner symbol c . Every digit matrix fixes the first coordinate vector, so the replacement of C makes these matrix comparisons equalities. The first coordinate of v_c cancels from their first-offset differences, and its second coordinate is retained as σ . Their first-offset gaps are therefore unchanged. The six swap comparisons are unchanged as well. This proves preservation of all weak rules and transfer of all eleven zero gaps back to the original boundaries.

For a digit map, its homogeneous matrix and its transformed matrix are

$$\mathcal{H}_s = \begin{pmatrix} 1 & q_s & h_s \\ 0 & z_s & k_s \\ 0 & 0 & 1 \end{pmatrix}, \quad \Phi(\mathcal{H}_s) = J\mathcal{H}_s^\top J = \begin{pmatrix} 1 & k_s & h_s \\ 0 & z_s & q_s \\ 0 & 0 & 1 \end{pmatrix}, \quad (36)$$

where J reverses the three homogeneous coordinates. Thus Φ exchanges q_s and k_s , preserves z_s and h_s , and again gives a nonnegative affine map with first diagonal entry one. The normalized boundary images can also be written explicitly:

$$\Phi(\mathcal{H}_{\widehat{C}}) = \begin{pmatrix} 1 & \sigma & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \Phi(\mathcal{H}_{\widehat{D}}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & \lambda \\ 0 & 0 & 1 \end{pmatrix}.$$

These are the forward normalized forms $C'(x, y) = (x + \sigma y, 0)$ and $D'(x, y) = (x, \lambda)$, respectively.

Since $J^2 = I$, the identity $\Phi(H_1 H_2) = \Phi(H_2) \Phi(H_1)$ reverses every interpreted word. Transposition and the coordinate permutation preserve entrywise order, and the matrix entry $(0, 2)$ is fixed by Φ . This entry is precisely the first affine offset. Consequently every weak reversed rule becomes the corresponding weak forward rule, with the same first-offset gap. The boundary *forms* exchange; the symbol labels do not. Proposition 19 therefore gives zero gaps for the reversed normalized model too. The normalization is essential here: it ensures that both transformed boundary matrices still represent affine maps.

Proof of Theorem 1. Lemma 3 puts the five digit matrices in one common triangular orientation, with unit first diagonal entries. For lower digits, Proposition 12 applies to the original admissible boundaries. For upper digits in the forward orientation, normalize the boundaries and apply Proposition 19. For upper digits in the reversed orientation, use (35) and Φ first, then apply the same proposition. In each case the normalization transfers all eleven zero first-offset gaps to the original interpretation. This is the conclusion of the theorem; it does not assert that every coefficient of every weak rule difference is zero. $\square$

7 The unrestricted reversed TOP obstruction

We now prove Theorem 2. Throughout this section, all coefficients are nonnegative real numbers, but no diagonal entry is assumed to be at least one. We distinguish the digit matrices A, B, E, F, G from their offsets a, b, e, f, g , and write $\mathcal{A}(x) = Ax + a$, $\mathcal{B}(x) = Bx + b$, and similarly for the ternary digits. Put $\gamma = v_c$. Fix an output coordinate i of F_d , and let r be row i of M_d . The two gaps in this coordinate are

$$ra, \quad r(b - g). \quad (37)$$

Here and below rows act on columns by juxtaposition. Both expressions are nonnegative by weak orientation. It suffices to derive a contradiction when at least one is positive. We also write $q = rB$.

For reference, the six reversed swap comparisons have matrix parts

$$AE \leq EA, \quad BE \leq FA, \quad AF \leq GA, \quad BF \leq EB, \quad AG \leq FB, \quad BG \leq GB. \quad (38)$$

Their offset parts are obtained by composing the corresponding affine maps. The boundary and readout comparisons used below are

$$r \leq rA, \quad rG \leq rB, \quad rg \leq rb, \quad (39)$$

$$B\gamma + b \leq E\gamma + e, \quad (40)$$

$$A^2\gamma + Aa + a \leq F\gamma + f, \quad (41)$$

$$BA\gamma + Ba + b \leq G\gamma + g. \quad (42)$$

All vector and matrix comparisons in this section are entrywise.

7.1 Arithmetic growth and two readout consequences

Define the binary interpretation $z(n)$, for positive integers n , by

$$z(1) = \gamma, \quad z(2n) = \mathcal{A}(z(n)), \quad z(2n+1) = \mathcal{B}(z(n)) \quad (n \geq 1).$$

The reversed swaps and the three c -boundary rules give

$$z(3n) \leq \mathcal{E}(z(n)), \quad z(3n+1) \leq \mathcal{F}(z(n)), \quad z(3n+2) \leq \mathcal{G}(z(n)) \quad (n \geq 1). \quad (43)$$

To see the induction explicitly, the case $n = 1$ consists of (40)–(42). For $n = 2m$, the three binary recursions, followed by the induction hypothesis, give $\mathcal{AE}$, $\mathcal{BE}$, and $\mathcal{AF}$; the first three reversed swaps bound these by $\mathcal{EA}$, $\mathcal{FA}$, and $\mathcal{GA}$. For $n = 2m+1$, the last three swaps similarly replace $\mathcal{BF}$, $\mathcal{AG}$, and $\mathcal{BG}$ by $\mathcal{EB}$, $\mathcal{FB}$, and $\mathcal{GB}$. Nonnegativity makes every digit map monotone, so these substitutions preserve the comparisons.

Lemma 20. *If one of the gaps in (37) is positive and $q = rB$, then $q\mathcal{A}^n(\gamma)$ tends to infinity as n tends to infinity. Moreover, qA is not a nonnegative scalar multiple of q .*

Proof. Let $R(n) = rz(n) + (v_d)_i$, and choose a positive δ no larger than one positive gap. Equations (39) and (43) imply

$$R(n) \leq R(2n), \quad R(3n+2) \leq R(2n+1).$$

One of these two families improves by at least δ for every $n \geq 1$. Combining one even step and one odd step gives

$$R(2^e 3t + 1) + \delta \leq R(2^{e+2}t + 1) \quad (e \geq 0, t \geq 1).$$

Iteration, using $R \geq 0$, yields $R(2^e t + 1) \geq k\delta$ whenever $e \geq 2k$. Since $z(2^{n+1} + 1) = \mathcal{BA}^n(\gamma)$, subtraction of the fixed number $rb + (v_d)_i$ proves the claimed divergence.

Suppose next that $qA = \theta q$ with $\theta \geq 0$. If $\theta < 1$, the scalar recurrence

$$q\mathcal{A}^{n+1}(\gamma) = \theta q\mathcal{A}^n(\gamma) + qa$$

is bounded by $q\gamma + qa/(1 - \theta)$, a contradiction. If $\theta \geq 1$, then $rG \leq q \leq qA$. Apply r to (42) and use $\gamma, a \geq 0$ and $rg \leq rb$. Every nonnegative difference must vanish, so

$$rg = rb, \quad qa = 0. \quad (44)$$

We record the short remaining argument because it will also be useful below. The matrix and offset parts of $ga \rightarrow af$ and $fa \rightarrow be$, together with (39) and (44), give

$$rGa = 0, \quad ra \leq rFa, \quad rF \leq rAF \leq rGA \leq rBA.$$

Consequently $0 \leq ra \leq qAa = \theta qa = 0$. Both gaps vanish, contrary to the assumption. $\square$

Lemma 21. *Let W be any word in the five digit maps and let M_W be its matrix. Under the strict-gap assumption,*

$$rM_W \mathcal{B}^n(\gamma) \longrightarrow \infty.$$

In particular, $M_W \neq 0$ for every such word.

Proof. Use the same function R and positive gap δ as in the preceding proof. Associate the integer maps $2n, 2n+1, 3n, 3n+1, 3n+2$ to a, b, e, f, g , respectively. Their composition for W is $sn + h$, where $s \geq 1$ and $0 \leq h < s$. Binary recursion and (43) imply

$$z(sn + h) \leq W(z(n)).$$

Since $z(2^{n+1} - 1) = \mathcal{B}^n(\gamma)$, it is enough to show that $R(s2^{n+1} - d)$ tends to infinity for the fixed positive integer $d = s - h$; subtracting the fixed offset of $F_d \circ W$ then gives the assertion about rM_W .

Here is an arithmetic proof of this assertion which uses only the two rank comparisons in Lemma 20. Consider the accelerated integer map

$$T(m) = \begin{cases} m/2, & m \text{ even}, \\ (3m+1)/2, & m \text{ odd}, \end{cases}$$

and stop a positive trajectory at 1. At every step above 1, R does not increase; it decreases by at least δ on every even step or on every odd step, according to which dynamic gap is positive.

If the even gap is positive, every trajectory must reach 1. Indeed, an infinite trajectory with infinitely many even steps would make R negative. An infinite final run of odd steps is also impossible: after j consecutive odd steps, $T^j(m) + 1 = 3^j(m+1)/2^j$, so $m+1$ would be divisible by arbitrarily large powers of two. If a terminating trajectory from m has k even steps, reversing it shows $m \leq 2^k$: a reversed even step doubles its argument, and a reversed odd step decreases it. Therefore $R(m) \geq k\delta$ and tends to infinity with m .

If the odd gap is positive, start the same arithmetic map at the negative integer $-d$. The trajectory stays negative. It has infinitely many odd steps, since a final run consisting only of halvings would require a nonzero integer to be divisible by every power of two. Fix a finite segment with at least k odd steps. For every sufficiently large e , the positive trajectory from $s2^e - d$ follows exactly the same parity pattern over this segment and remains above 1. More explicitly, at step j the difference between the positive and negative trajectories is $s3^{o_j}2^{e-j}$, where o_j counts the preceding odd steps. Taking e larger than the segment length preserves parity, and taking it still larger keeps all these positive states above 1. Hence $R(s2^e - d) \geq k\delta$ for all sufficiently large e . Since k was arbitrary, the required divergence follows. $\square$

The sum of the six nonnegative differences in (38) has trace zero, by cyclicity of the trace. Every diagonal entry of every one of those differences is therefore zero. We use these six diagonal equalities even when the full swaps are not exact.

Lemma 22 (The binary products cannot be ordered). *Under the strict-gap assumption, $AB \not\leq BA$.*

Proof. Suppose $AB \leq BA$. The readout inequalities give

$$rG \leq q = rB \leq rAB \leq rBA = qA.$$

Applying r to (42), each term in

$$(qA - rG)\gamma + qa + (rb - rg) \leq 0$$

is nonnegative. Thus $qa = 0$ and $rb = rg$. The row q itself satisfies the readout conditions: $q \leq qA$, $qG \leq qB$, and $qg \leq qb$. Indeed, $BG \leq GB$ gives $qG \leq rGB \leq qB$, while its offset comparison $Bg + b \leq Gb + g$ gives $qg \leq rGb + rg - rb \leq qb$. Repeating the preceding argument with q in place of r gives $qBa = 0$. As in the proof of Lemma 20, the ga, fa comparisons also give

$$ra \leq rFa \leq qAa. \tag{45}$$

Since $rb = rg$, strictness requires $ra > 0$.

Choose a coordinate i with $r_ia_i > 0$, and let j be the other coordinate. The equality $qa = 0$ gives $q_i = 0$. The row q is nonzero by Lemma 20, so $q_j > 0$ and $a_j = 0$. From $q_i = (rB)_i = 0$ and $qBa = 0$, respectively, we obtain $B_{ii} = B_{ji} = 0$. The positive rightmost expression in (45)

gives $A_{ji} > 0$. The (j, j) entry of $AB \leq BA$ then gives $B_{ij} = 0$. Consequently $q_j = r_j B_{jj} > 0$, so $r_j, B_{jj} > 0$. The (i, j) entry of the same comparison gives $A_{ij} = 0$, and $r \leq rA$ in coordinate j gives $A_{jj} \geq 1$. In the coordinate order (i, j) , write

$$A = \begin{pmatrix} u & 0 \\ v & t \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & s \end{pmatrix}, \quad v > 0, \quad t \geq 1, \quad s > 0, \quad a = (a_i, 0)^\top, \quad a_i > 0.$$

We now spell out the swap entries that determine the remaining shape. The diagonal equality $(FA)_{ii} = (BE)_{ii} = 0$ gives $F_{ij} = 0$ and $F_{ii}u = 0$. The (j, i) comparison $BF \leq EB$ gives $F_{ji} = 0$. The same entry of $AG \leq FB$ gives $G_{ii} = G_{ji} = 0$; then $(GA)_{ii} = (AF)_{ii} = F_{ii}u = 0$ gives $G_{ij} = 0$. The equality $(EA)_{ii} = (AE)_{ii}$ gives $E_{ij} = 0$. The diagonal equalities for EB, BF and for GA, AF now imply

$$E_{jj} = F_{jj} = G_{jj} =: \nu.$$

Since $r_i, a_i > 0$ and $F_{ji} = 0$, the first inequality in (45) gives $F_{ii} \geq 1$. Thus $u = 0$. The (j, i) comparison $AF \leq GA$ gives $vF_{ii} \leq \nu v$, so $\nu > 0$. Finally $(FA)_{jj} = (BE)_{jj}$ gives $\nu t = s\nu$. Hence $s = t =: \lambda \geq 1$, and $rG \leq rB$ gives $\nu \leq \lambda$. The resulting shapes are

$$\begin{aligned} A &= \begin{pmatrix} 0 & 0 \\ v & \lambda \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & \lambda \end{pmatrix}, \quad E = \begin{pmatrix} \xi & 0 \\ \eta & \nu \end{pmatrix}, \\ F &= \begin{pmatrix} \rho & 0 \\ 0 & \nu \end{pmatrix}, \quad G = \begin{pmatrix} 0 & 0 \\ 0 & \nu \end{pmatrix}, \quad v > 0, \quad 1 \leq \rho \leq \nu \leq \lambda. \end{aligned} \tag{46}$$

Equation (42) now gives

$$b_i \leq g_i, \quad \lambda v \gamma_i + (\lambda^2 - \nu) \gamma_j + b_j \leq g_j.$$

Both extra terms are nonnegative. Since $r_i, r_j > 0$ and $rb = rg$, we obtain $g = b$ and $\gamma_i = 0$. The coordinate- j offset comparisons for fa and ga give

$$\lambda e_j + b_j \leq f_j, \quad v f_i + \lambda f_j \leq b_j.$$

Because $\lambda \geq 1$, these imply $f_j = b_j$ and $e_j = 0$. The ec comparison, using $\gamma_i = e_j = 0$, becomes

$$\lambda \gamma_j + b_j \leq \nu \gamma_j,$$

so $b_j = f_j = g_j = 0$. The coordinate- j comparison for $fb \rightarrow ag$ consequently gives $v g_i \leq 0$, hence $g_i = 0$. Finally the coordinate- i comparison for $ga \rightarrow af$ says $a_i \leq g_i = 0$, contradicting $a_i > 0$. $\square$

7.2 Excluding support in both directions

Put $H = A + B + E + F + G$. In two coordinates, every coordinate can return to every other through a product of digit matrices exactly when $H_{01}, H_{10} > 0$. Indeed, these entries themselves give the two directions; if either entry of H is zero, the same entry is zero in every power of H . We show that the case

$$H_{01} > 0, \quad H_{10} > 0 \tag{47}$$

is incompatible with a strict dynamic gap.

First all six matrix swaps are exact in this case. Let $S_0 = A + B$, $T_0 = E + F + G$, and $K = T_0 S_0 - S_0 T_0 \geq 0$, the sum of the six swap differences. Its diagonal entries are zero. Cyclicity of the trace gives $\text{tr}(HK) = 0$, since $H = S_0 + T_0$. But

$$\text{tr}(HK) = H_{10}K_{01} + H_{01}K_{10}.$$

Under (47), both remaining entries of K vanish. Every nonnegative summand in K therefore vanishes, proving exactness. Also $AB \neq BA$ by Lemma 22.

Lemma 23. *Under (47) and the strict-gap assumption, F is invertible.*

Proof. Suppose $\det F = 0$. Lemma 21 gives $F^2 \neq 0$, so Cayley–Hamilton gives

$$\tau = \operatorname{tr} F > 0, \quad F^2 = \tau F.$$

Exact swaps also give

$$F(AB) = B^2 F, \quad F(BA) = A^2 F. \quad (48)$$

Assume first that $\operatorname{tr} B > 0$. From the first identity, $F(B^2 F) = \tau B^2 F$. Hence B^2 preserves the one-dimensional image of F , which is the τ -eigenspace of F . Cayley–Hamilton for B , with $\operatorname{tr} B \neq 0$, shows that B also preserves this line. Thus $BF = b_* F$ for a scalar b_* . Lemma 21 gives $BF \neq 0$, so $b_* \neq 0$. Since $EB = BF$, multiplying on the right by F and cancelling b_* gives $EF = \tau F$.

If E is nonscalar, its centralizer in dimension two consists of $xI + yE$, as proved in Section 3. Since $EA = AE$, we get $A = xI + yE$ and therefore $AF = (x + y\tau)F$. If E is scalar, $EF = \tau F$ makes $E = \tau I$ and $EB = BF$ gives $B = (b_*/\tau)F$. The equality $GB = BG$ then gives $GF = FG$, while $AG = FB = b_* F$ and $GA = AF$ imply

$$b_* GF = G(AG) = (GA)G = A(FG) = (AG)F = b_* \tau F.$$

Thus $GF = \tau F$, and $\tau AF = A(GF) = (AG)F = b_* \tau F$. In either case there is a scalar a_* with $AF = a_* F$.

Put $J = AB - BA$. These two scalar actions give $JF = 0$ and $\operatorname{tr} J = 0$. The polarized Cayley–Hamilton identity gives $FJ = \tau J$, while (48) gives $FJ = (b_*^2 - a_*^2)F$. Taking traces yields $(b_*^2 - a_*^2)\tau = 0$; consequently $J = 0$. This contradicts Lemma 22. If $\operatorname{tr} B = 0$ but $\operatorname{tr} A > 0$, interchange A, B and E, G in this argument.

Finally suppose both binary traces are zero. Nonnegativity and $A^2, B^2 \neq 0$ give

$$A = \begin{pmatrix} 0 & a_* \\ b_* & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & c_* \\ d_* & 0 \end{pmatrix}, \quad a_*, b_*, c_*, d_* > 0.$$

The commuting endpoint swaps force $E_{00} = E_{11} = e_*$ and $G_{00} = G_{11} = g_*$. The off-diagonal entries of $FA = BE$ and $GA = AF$ give

$$F_{00} = (c_*/a_*)e_*, \quad F_{11} = (d_*/b_*)e_*, \quad F_{00} = F_{11} = g_*.$$

Since $\operatorname{tr} F > 0$, $e_* > 0$, so $a_* d_* = b_* c_*$ and again $AB = BA$. This completes all trace cases. $\square$

We next exclude singular binary matrices. Determinants in the exact swaps give $\det A = \det B$ because F is invertible. Suppose this common determinant is zero. The nonzero products A^2, B^2 force positive traces. Taking traces in (48), after conjugation by F , gives

$$\operatorname{tr} A = \operatorname{tr} B = \alpha > 0, \quad \operatorname{tr}(AB) = \alpha^2.$$

For $N = B - A \neq 0$, we obtain $\operatorname{tr} N = \det N = \operatorname{tr}(AN) = 0$. The nilpotent-line argument from Lemma 7 gives

$$AN = sN, \quad NA = (\alpha - s)N, \quad s \in \{0, \alpha\}.$$

If $s = 0$, then $BA = \alpha B$, so $qA = \alpha q$, contrary to Lemma 20. If $s = \alpha$, the remaining alternative is

$$\begin{aligned} A^2 &= BA = \alpha A, & AB &= B^2 = \alpha B, \\ F &= \beta I, & E &= (\beta/\alpha)A, & G &= (\beta/\alpha)B, & \alpha, \beta &> 0. \end{aligned} \quad (49)$$

Here is why the scalar F and the endpoint formulas follow without assuming $\operatorname{tr} F > 0$ in advance. The two intertwining identities now say that F commutes with both A and B . Since A is

nonscalar, write $F = xI + yA$; commutation with B and $AB \neq BA$ force $y = 0$. Invertibility and nonnegativity give $x = \beta > 0$. Writing $E = x'I + y'A$ and using $FA = BE$ determines $E = (\beta/\alpha)A$; the same argument gives G .

The off-diagonal entries of H are positive multiples of those of $A + B$. Thus $u = (A + B)(1, 1)^\top$ is strictly positive. Equation (49) gives $Au = Bu = \alpha u$ and $Gu = \beta u$. The nonzero row r satisfies $ru > 0$; dotting the readout comparisons with u gives

$$1 \leq \alpha, \quad 0 < \beta \leq \alpha. \quad (50)$$

We give the offset elimination for this projection case explicitly. If $\alpha > 1$, put $p = rA$ and $s = rB$, and abbreviate

$$\begin{aligned} x &= p\gamma, \quad y = s\gamma, \quad u_0 = pa, \quad u_1 = sa, \\ b_0 &= pb, \quad b_1 = sb, \quad g_0 = pg, \quad g_1 = sg, \quad e_0 = pe, \quad f_0 = pf. \end{aligned}$$

These symbols denote scalar projections only within this paragraph. The offset comparisons for $ea, ec, gc, fc, ga, fb, gb$, projected with p, p, s, p, p, p, s , respectively, give

$$\begin{aligned} \alpha e_0 + u_0 &\leq \beta u_0 + e_0, & \alpha y + b_0 &\leq \beta x + e_0, \\ \alpha^2 x + \alpha u_1 + b_1 &\leq \beta y + g_1, & \alpha^2 x + (\alpha + 1)u_0 &\leq \beta x + f_0, \\ \alpha f_0 + u_0 &\leq \beta u_1 + g_0, & \alpha g_0 + u_0 &\leq \beta b_0 + f_0, \\ \alpha g_1 + b_1 &\leq \beta b_1 + g_1. \end{aligned}$$

The first and last imply $e_0 \leq u_0$ and $g_1 \leq b_1$. Replace e_0 by u_0 in the second inequality, then add the second, third, fifth, sixth, and fourth inequalities with respective weights $\beta, \alpha, \alpha, 1, \alpha^2 - 1$. Cancellation gives

$$\begin{aligned} [(\alpha^2 - 1)(\alpha^2 - \beta) + (\alpha^3 - \beta^2)]x + [\alpha^2(\alpha + 1) - \beta]u_0 \\ + \alpha(\alpha - \beta)u_1 + \alpha(b_1 - g_1) \leq 0. \end{aligned}$$

Every term is nonnegative, and the coefficients of x, u_0 are strictly positive by (50). Thus $x = u_0 = 0$, and $\alpha y + b_0 \leq \beta x + u_0$ gives $b_0 = 0$. Since $r \leq p$, we obtain $ra = rb = 0$, and then $rg = 0$.

If $\alpha = 1$, positivity of u and $r \leq rA$ imply $rA = r$: the nonnegative row $rA - r$ has zero product with $u > 0$. With $q = rB$, the projection identities give

$$qA = r, \quad qB = q, \quad rF = \beta r, \quad rG = \beta q, \quad qE = \beta r.$$

Project the offset comparisons for fa, ga, gc, ec by r, r, r, q and give them weights $\beta, \beta, 1, \beta$. After cancellation their sum is

$$\beta(1 - \beta)ra + (1 - \beta^2)qa + (1 - \beta^2)r\gamma + (1 + \beta)(rb - rg) + \beta qb \leq 0.$$

All terms are nonnegative. Since $\beta > 0$, $qb = 0$. Together with $qB = q$, this makes $q\mathcal{B}^n(\gamma) = q\gamma$ for every n , contradicting Lemma 21 with the one-letter prefix b . Thus the singular-binary case is impossible.

We have proved that A, B, F are invertible. The exact swaps then make E, G invertible as well. We check the trace hypotheses needed to reuse the nonsingular classification from Section 3. Equation (48) gives $\text{tr } A = \text{tr } B \geq 0$ as before. If both traces were zero, Cayley–Hamilton and $\det A = \det B$ would give $A^2 = B^2$; cancellation of F in (48) would then give $AB = BA$. Hence both binary traces are positive. The ternary matrices have the same trace because $BF = EB$ and $AF = GA$ are invertible intertwinings. If that trace were zero, nonnegativity would make all their diagonal entries zero. The equality $BG = GB$, with G invertible, would force $B_{00} = B_{11} > 0$. The off-diagonal entries of $BF = EB$ would then give $E = F$; $FA = BE$ and $EA = AE$ would

give $FA = BF = AF$, and cancellation would force $A = B$. This again contradicts Lemma 22. Thus the ternary traces are positive.

Transpose all five matrices and apply Lemma 8. Transposing back gives

$$\begin{aligned} \mu, t > 0, \quad \text{tr } A = \text{tr } B = 3\mu, \quad \det A = \det B = 2\mu^2, \\ A(B - A) = 2\mu(B - A), \quad (B - A)A = \mu(B - A), \\ E = t(2A - \mu I), \quad F = t(A + B - \mu I), \quad G = t(2B - \mu I). \end{aligned} \quad (51)$$

Put $U = A - \mu I$, $V = B - \mu I$, and $S = U + V$. The trace and determinant equations make U, V nonnegative, by the same diagonal calculation as in Section 3. The product identities are now the transposes of (15):

$$U^2 = VU = \mu U, \quad UV = V^2 = \mu V, \quad US = VS = \mu S, \quad S^2 = 2\mu S.$$

Since $H_{ij} = (1 + 3t)S_{ij}$ for $i \neq j$, both off-diagonal entries of S are positive. Therefore

$$u = S(1, 1)^T > 0, \quad Au = Bu = \lambda u, \quad Eu = Fu = Gu = \nu u, \quad \lambda = 2\mu, \quad \nu = 3t\mu.$$

Dotting (39) with u gives $1 \leq \lambda$ and $0 < \nu \leq \lambda$.

The case $\lambda = 1$ needs a separate argument. The positive row $w = (1, 1)S$ satisfies $w(A + B) = 4\mu w$ and $w(E + F + G) = 9t\mu w$. Summing the six offset comparisons gives

$$3(a + b) + (A + B)(e + f + g) \leq (E + F + G)(a + b) + 2(e + f + g).$$

When $\lambda = 1$, $\mu = 1/2$, and projection by w reduces this to $3(1 - \nu)w(a + b) \leq 0$. If $\nu < 1$, positivity of w implies $a = b = 0$, so neither gap is positive. If $\nu = 1$, then $t = 2/3$. The positive vector u and $r \leq rA$ give $rA = r$; likewise $rG \leq rB$ and their equal products with u give $rG = rB$. Substitution of $G = (2/3)(2B - I/2)$ gives $rB = r$. Thus $qA = q$, contrary to Lemma 20.

It remains to exclude $\lambda > 1$ in this support case. All diagonal entries of A, B are at least $\mu > 0$, and both off-diagonal entries of $A + B$ are positive. At least one of a, b is nonzero. Apply its digit to γ , obtaining a nonzero nonnegative vector, then apply a binary digit whose positive off-diagonal transfers any missing coordinate. Its positive diagonal preserves the already positive coordinate. Thus some $p_0 \in \{4, 5, 6, 7\}$ has $z(p_0) > 0$ coordinatewise, and hence $z(p_0) \geq cu$ for a number $c > 0$. Appending k zero digits yields $z(p_0 2^k) \geq c\lambda^k u$.

Choose $K, M \geq 0$ with $\gamma \leq Ku$ and $e, f, g \leq Mu$. Because all three ternary matrices send u to $\nu u \leq \lambda u$, the conversion induction gives

$$z(m) \leq (K + jM)\lambda^j u \quad (3^j \leq m < 2 \cdot 3^j).$$

For each n , choose k with $3^{2(n+1)} \leq p_0 2^k < 2 \cdot 3^{2(n+1)}$. Taking the least possible k ensures the upper bound, and $p_0 \leq 7$ gives $k \geq 3n$ as in the elementary comparison below (60). Since $u > 0$, comparison of the two vector bounds gives

$$c\lambda^{3n} \leq (K + 2(n+1)M)\lambda^{2(n+1)}, \quad \text{hence} \quad c\lambda^n \leq \lambda^2(K + 2(n+1)M).$$

This is impossible for $\lambda > 1$, since an exponential cannot be bounded by a linear function. Thus (47) is excluded.

Consequently $H_{01} = 0$ or $H_{10} = 0$. Since every digit matrix is nonnegative, all five share the corresponding zero entry. This proves the required common triangular orientation.

7.3 The active triangular coordinate

It remains to consider a common triangular orientation of the five digit matrices. We may take it to be upper triangular. This coordinate choice requires some care because the output coordinate

of F_d has already been fixed. If the common orientation is lower triangular, put $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and replace F_s by $P \circ F_s \circ P$ for $s \in \{a, b, c, e, f, g\}$, but replace F_d by $F_d \circ P$. Thus M_d changes to $M_d P$ and v_d stays unchanged. Every reversed word comparison is preserved; the two offset vectors $v_{da} - v_d$ and $v_{db} - v_{dg}$ are unchanged. In particular, the chosen strict output coordinate is preserved. This is an input coordinate change at d , not a permutation of its output.

Write

$$A = \begin{pmatrix} x & p \\ 0 & \alpha \end{pmatrix}, \quad B = \begin{pmatrix} y & p_b \\ 0 & \beta \end{pmatrix}. \quad (52)$$

As before, $q = rB$ denotes the readout row. Lemma 20 shows that $q_0 > 0$: if $q_0 = 0$, then $qA = \alpha q$. Hence $r_0 > 0$ and $y > 0$. The first coordinates of $r \leq rA$ and $rG \leq rB$ now give

$$x \geq 1, \quad y > 0, \quad G_{00} \leq y. \quad (53)$$

Lemma 22 also gives

$$(BA)_{01} < (AB)_{01}. \quad (54)$$

Indeed, the diagonal entries of AB and BA agree, and their lower left entries are zero.

Lemma 24. *Under the strict-gap assumption, $\alpha < 1$.*

Proof. If $\alpha \geq 1$, then $I \leq A$. Thus $q \leq qA$, and the proof of (44) again gives $rg = rb$ and $qa = 0$. Since $q_0 > 0$, we have $a_0 = 0$. If $r_1 = 0$, this already gives $ra = 0$. Otherwise, if $\beta > 0$, then $q_1 = r_0 p_b + r_1 \beta > 0$, and $qa = 0$ gives $a_1 = 0$. If $\beta = 0$, (54) reads $yp + \alpha p_b < xp_b$ and forces $p_b > 0$; again $q_1 > 0$ and $a_1 = 0$. Both gaps vanish in every case. $\square$

7.4 A zero active ternary diagonal

We first dispose of $F_{00} = 0$. The $(0,0)$ entries of $BE \leq FA$ and $AG \leq FB$, with (53), force $E_{00} = G_{00} = 0$. Put $u = E_{01}$, $v = F_{01}$, $w = G_{01}$ and $\epsilon = E_{11}$, $\phi = F_{11}$, $\psi = G_{11}$. The relevant upper right comparisons are

$$\begin{aligned} xu + p\epsilon &\leq \alpha u, & yu + p_b \epsilon &\leq \alpha v, & xv + p\phi &\leq \alpha w, \\ yv + p_b \phi &\leq \beta u, & xw + p\psi &\leq \beta v, & yw + p_b \psi &\leq \beta w. \end{aligned}$$

If $v > 0$, the fourth comparison forces $u > 0$ and $\beta > 0$. The first then gives $\alpha \geq x \geq 1$, contradicting Lemma 24. Thus $v = 0$. The second and fifth comparisons imply $u = w = 0$. Equation (54) implies $p + p_b > 0$; the third and fourth comparisons therefore give $\phi = 0$, and the fifth and sixth give $\psi = 0$. Hence $F = G = 0$.

The offset parts of $fa \rightarrow be$ and $ga \rightarrow af$ now read $Be + b \leq f$ and $Af + a \leq g$. After applying r and using $r \leq rA$ and $rg \leq rb$, we obtain

$$rb \leq rBe + rb \leq rf \leq rAf \leq rAf + ra \leq rg \leq rb.$$

All comparisons are equalities. In particular, $ra = 0$ and $rb = rg$, which excludes the zero active ternary diagonal.

7.5 Positive active diagonals and the unit cases

Suppose $F_{00} > 0$. The diagonal equalities in the six swaps give

$$A_{00} = B_{00} = \lambda \geq 1, \quad E_{00} = F_{00} = G_{00} = \nu, \quad 0 < \nu \leq \lambda. \quad (55)$$

For completeness, BF and EB have equal $(0,0)$ entries, so $E_{00} = F_{00}$ because $B_{00} > 0$. Equality for AF and GA gives $G_{00} = F_{00}$; equality for BE and FA then gives $A_{00} = B_{00}$. The last bound is (53).

If $\lambda = \nu = 1$, the previously proved admissible obstruction applies after a change of the two boundary maps. Retain all five digit maps and replace the boundaries by

$$\widehat{F}_c(x) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} x + \gamma, \quad \widehat{F}_d(x) = \begin{pmatrix} 1 & r_1/r_0 \\ 0 & 0 \end{pmatrix} x.$$

Every digit matrix X satisfies $X \operatorname{diag}(1, 0) = \operatorname{diag}(1, 0)$. The three c -boundary matrix comparisons are therefore equalities, and their offsets are the original offsets. The two d -boundary comparisons follow by dividing the selected original readout row by r_0 ; their second output coordinates are zero. All seven new matrices have first diagonal entry one. Thus Theorem 1, already proved in Sections 3–6, applies to this reversed admissible interpretation. Its two dynamic gaps are the original gaps divided by $r_0 > 0$, a contradiction.

Suppose instead that $\lambda = 1$ and $0 < \nu < 1$. Lemma 24 gives $0 \leq \alpha < 1$. Define the nonnegative row

$$w = \left(1, \frac{p}{1-\alpha}\right), \quad wA = w.$$

From $AE \leq EA$, its upper right entry gives $E_{01} + pE_{11} \leq \nu p + \alpha E_{01}$, hence $wE \leq \nu w$. The corresponding offset comparison is $Ae + a \leq Ea + e$. Apply w and cancel we to obtain

$$0 \leq wa \leq wEa \leq \nu wa.$$

Thus $wa = 0$ and $w\mathcal{A}^n(\gamma) = w\gamma$ for every n . The second coordinate of the same orbit obeys the bounded recurrence $t_{n+1} = \alpha t_n + a_1$, with $t_n \leq \gamma_1 + a_1/(1-\alpha)$. The first coordinate is at most $w\gamma$ because $w_0 = 1$. Consequently the entire orbit, and therefore its value under the nonnegative row $q = rB$, is bounded. This contradicts Lemma 20.

7.6 The expanding active diagonal

Only $\lambda > 1$ remains. Keep

$$A = \begin{pmatrix} \lambda & p \\ 0 & \alpha \end{pmatrix}, \quad B = \begin{pmatrix} \lambda & p_b \\ 0 & \beta \end{pmatrix}, \quad 0 \leq \alpha < 1.$$

We first eliminate $\beta > 1$, including cases where the other ternary diagonal entries vanish. Equality of the diagonal swap entries gives $E_{11} = F_{11}$ and $\beta E_{11} = \alpha F_{11}$. Hence $E_{11} = F_{11} = 0$. The second-coordinate offset comparisons of $fa \rightarrow be$ and $eb \rightarrow bf$ become

$$\beta e_1 + b_1 \leq f_1, \quad \beta f_1 + b_1 \leq e_1.$$

Since $\beta > 1$, they force $e_1 = f_1 = b_1 = 0$. The second coordinate of $ea \rightarrow ae$ then gives $a_1 = 0$, and (40) gives $\gamma_1 = 0$.

The first-coordinate offset comparison of $gb \rightarrow bg$ is

$$\lambda g_0 + p_b g_1 + b_0 \leq \nu b_0 + g_0.$$

Since $\nu \leq \lambda$ and $\lambda > 1$, it implies $g_0 \leq b_0$. Equation (42) reduces in its first coordinate to

$$(\lambda^2 - \nu)\gamma_0 + \lambda a_0 + b_0 \leq g_0.$$

Both coefficients on the left are positive, so $\gamma_0 = a_0 = 0$. The first coordinate of $ea \rightarrow ae$ then gives $(\lambda - 1)e_0 \leq 0$, so $e_0 = 0$; finally (40) gives $b_0 = 0$. We have $a = b = 0$, which precludes a positive gap because $ra = 0$ and $0 \leq r(b - g) = -rg \leq 0$.

We may therefore assume $0 \leq \beta \leq 1$. Equation (54) becomes

$$\Delta = (\lambda - \alpha)p_b - (\lambda - \beta)p > 0. \tag{56}$$

In particular $p_b > 0$. Write

$$E = \begin{pmatrix} \nu & u \\ 0 & \epsilon \end{pmatrix}, \quad F = \begin{pmatrix} \nu & v \\ 0 & \phi \end{pmatrix}, \quad G = \begin{pmatrix} \nu & w \\ 0 & \psi \end{pmatrix}.$$

We next show $\epsilon, \phi, \psi \leq \nu$. The upper right comparison for $gb \rightarrow bg$ reads

$$(\lambda - \beta)w + p_b(\psi - \nu) \leq 0,$$

so $\psi \leq \nu$. If $\epsilon > \nu$, the comparison for $ea \rightarrow ae$, $(\lambda - \alpha)u + p(\epsilon - \nu) \leq 0$, forces $u = p = 0$. The comparisons for $fa \rightarrow be$ and $eb \rightarrow bf$ would then give $p_b\epsilon \leq \alpha v$ and $\lambda v + p_b\phi \leq \nu p_b$. Thus $\lambda\epsilon \leq \alpha\nu$, contradicting $\epsilon > \nu$ and $\alpha < \lambda$. Hence $\epsilon \leq \nu$.

If $\alpha > 0$, equality of the lower diagonal entries of AF and GA gives $\phi = \psi \leq \nu$. If $\alpha = 0$ and $\phi > \nu$, the upper right comparison for $ga \rightarrow af$, $\lambda v + p(\phi - \nu) \leq 0$, forces $v = p = 0$. The comparison for $ea \rightarrow ae$ then gives $u = 0$, and that for $eb \rightarrow bf$ gives $p_b\phi \leq \nu p_b$, again a contradiction. Therefore

$$\epsilon, \phi, \psi \leq \nu \leq \lambda. \quad (57)$$

Although the three upper right entries need not be bounded by λ , all three ternary matrices admit a common positive upper bound in a suitably weighted coordinate system. Set

$$L = \frac{3\lambda - 1}{2}, \quad t = \frac{L - \lambda}{1 + u + v + w}, \quad h = (1, t)^\top. \quad (58)$$

Then $t > 0$, $L > \lambda > 1$, and

$$L^2 < \lambda^3, \quad \lambda^3 - L^2 = \frac{(\lambda - 1)^2(4\lambda - 1)}{4}.$$

Equation (57) gives $Eh, Fh, Gh \leq Lh$: for the first coordinate, each of ut, vt, wt is at most $L - \lambda$; for the second coordinate, each lower diagonal is at most $\lambda < L$.

At least one of a, b is nonzero under the strict-gap assumption. Starting from γ , apply that digit once to obtain a nonzero nonnegative vector. Since $A_{00} = \lambda > 0$ and $p + p_b > 0$, one further application of $\mathcal{A}$ or $\mathcal{B}$ produces a positive first coordinate. Thus there is a binary prefix $p_0 \in \{4, 5, 6, 7\}$ with $z(p_0)_0 = c > 0$. Appending k binary zero digits gives

$$z(p_0 2^k)_0 \geq c\lambda^k. \quad (59)$$

On the other hand, since h is strictly positive, choose $K, M \geq 0$ with $\gamma \leq Kh$ and $e, f, g \leq Mh$. Repeated application of (43) yields

$$z(m) \leq (K + jM)L^j h \quad (3^j \leq m < 2 \cdot 3^j). \quad (60)$$

The induction removes the last ternary digit of m . Its scalar estimate is $L(K + jM)L^j + M \leq (K + (j + 1)M)L^{j+1}$, using $L \geq 1$.

For each $n \geq 0$, choose the least k for which $3^{2(n+1)} \leq p_0 2^k$. Minimality gives $p_0 2^k < 2 \cdot 3^{2(n+1)}$, and $k \geq 3n$ because

$$p_0 2^{3n} \leq 7 \cdot 8^n < 9 \cdot 9^n = 3^{2(n+1)}.$$

Combine (59) and (60), using $h_0 = 1$, to obtain

$$c\lambda^{3n} \leq (K + 2(n + 1)M)L^{2(n+1)}.$$

This would bound $(\lambda^3/L^2)^n$ by a linear function of n , although $\lambda^3/L^2 > 1$. The contradiction is elementary: if $\rho = 1 + \eta > 1$, the binomial estimate $\rho^n \geq \binom{n}{2}\eta^2$ already exceeds every fixed linear bound for sufficiently large n .

Conclusion of the proof of Theorem 2. If one coordinate of either nonnegative dynamic gap vector were positive, the preceding argument would first force a common triangular orientation and would then exclude $F_{00} = 0$, $\lambda = \nu = 1$, $\lambda = 1 > \nu$, and $\lambda > 1$. These cases are exhaustive by (55). Hence every coordinate of both gap vectors is zero: $v_{da} = v_d$ and $v_{db} = v_{dg}$. No positive uniform decrement is therefore available for either reversed TOP rule in two nonnegative real affine coordinates. The conclusion concerns these two offset vectors. It does not assert that all eleven offset gaps vanish without the diagonal assumptions, and it does not settle the unrestricted forward TOP problem. $\square$

8 Formal statement, verification, and scope

The formalization uses Lean 4 and mathlib [3, 4]. Matrices and offsets are functions on `Fin 2` with values in $\mathbb{R}$. The shared definition `Affine.comp X Y` is

$$(M_X M_Y, M_X v_Y + v_X),$$

so the formal composition convention is the one used throughout this paper. `Admissible` is exactly (2); `ForwardWeak` and `ReversedRealWeak` each contain eleven coefficientwise comparisons. The two zero-gap records each contain eleven equalities at offset index zero.

The main declarations in `FullTwoCoordinate.lean` are

Declaration in <code>CollatzResearch.FullTwo</code>	Conclusion
<code>forward_all_gaps_zero</code>	Eleven forward gaps vanish
<code>reversed_all_gaps_zero</code>	Eleven reversed gaps vanish
<code>full_two_coordinate_obstruction</code>	Both implications

The unrestricted reversed result is in the file and namespace

`CollatzReversedRealTwoCoordinate.lean`
`CollatzResearch.RealTwoCoordinate.`

It assumes seven nonnegative affine maps and the eleven comparisons in `ReversedRealWeak`. Its two concluding declarations are

Declaration	Conclusion
<code>reversed_eligible_offsets_equal</code>	Both eligible offset differences are zero
<code>no_positive_eligible_offset_gap</code>	Neither eligible rule has a positive fixed gap in any coordinate

Each declaration reports only `propext`, `Classical.choice`, and `Quot.sound`. No additional axiom or proof placeholder is used. The combined dependency closure of both obstructions, including the soundness lemmas below, contains 120 local modules. These were rebuilt in a fresh directory, and an additional audit queries the axioms of all 515 public theorems and lemmas and 69 supporting definitions: 584 declarations in total. The retained report records the exact source inventory, source and compiled-module hashes, compiler outputs, dependency revisions, and the final theorem audit. Existing local compiled files are excluded from that fresh build.

The fixed-gap soundness lemmas use the namespace `CollatzResearch.FullTwoSoundness`:

<code>admissible_preserves_gap</code>	Preservation under admissible maps
<code>weak_rule_gives_gap</code>	Evaluation of a weak rule with an offset gap
<code>gap_wellFounded</code>	Well-foundedness for a positive decrement

These three declarations are included in the combined fresh build and axiom audit. They do not formalize the prior Collatz equivalence or the general rule-removal theorem.

The checked environment is Lean 4.27.0 with mathlib revision

a3a10db0e9d66acbebf76c5e6a135066525ac900.

The source for Theorem 1 has SHA-256

429785555eecec277248380fe0c2e767c8c7303cf0e2712de29878dda2bd29a0.

The source for Theorem 2 has SHA-256

b7cc9035e073d624fd270f6fbde9f2149a290dd93f238643a65aefe4fe434b71.

From the artifact root, the verification entry point is

```
python3 verify.py \
  --mathlib-root /path/to/pinned/mathlib \
  --build-root /path/to/temporary/builds \
  --report /path/to/fresh-check.json
```

The retained verification record is `verification/rebuild.json`. The claim-to-declaration map in `paper/lean-claims.md` identifies the supporting formal results. Appendix A exposes the actual definitions and theorem types, so the connection between the prose and the formal claim can be inspected directly.

The trust boundary consists of Lean’s kernel and the stated foundational axioms, together with the definitions shown in the appendix. Automated tactics construct proof terms that the kernel checks. The build rejects proof placeholders and additional axioms; it does not use numerical search results as proof. Kernel acceptance establishes the Lean proposition. The symbol table, appendix, and claim-to-declaration map address the separate question of whether that proposition is the one claimed in the paper. The Collatz equivalence is cited from prior work and is not asserted to have been re-formalized by this artifact.

8.1 Availability and reproducibility

The paper, Lean sources, verification script, retained compiler output, and claim-to-declaration map are available at

<https://github.com/x1xhlol/collatz-two-coordinate-obstruction>.

The repository records the source hashes and pins the dependencies. Reproduction uses a new output path to preserve the supplied audit record. Neither obstruction has a coefficient cap, integrality assumption, or nonsingularity assumption. Theorem 1 retains the diagonal bounds; Theorem 2 removes them for the two reversed eligible rules. Neither theorem proves the Collatz conjecture or resolves forward TOP removal in this dimension.

Acknowledgments

I thank Emre Yolcu, Scott Aaronson, and Marijn J. H. Heule for making their rewriting system and implementation available. The formalization uses Lean and mathlib, and I thank their developers and contributors.

Use of AI

OpenAI Codex was used extensively for mathematical exploration, development of the Lean proofs and verification scripts, and preparation and revision of this manuscript. The proof claims are supported by the accompanying Lean artifacts, rather than by the system’s assertions. This disclosure does not imply review or endorsement by the authors of the underlying rewriting system.

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A Exact Lean declarations

The blocks below reproduce the checked source verbatim, with the original line numbers. The theorem blocks end at their types; their proof terms are omitted. Source paths are relative to the repository root.

The affine definitions and `ReversedRealWeak` are in `CollatzCertificate`. The definitions and theorem with diagonal bounds are in `CollatzResearch.FullTwo`. Matrix notation is opened in both namespaces. The order on vectors is pointwise, so `Weak X Y` means that every coefficient of `Y` is at most the corresponding coefficient of `X`. Composition places the left argument outside the right argument. The unrestricted reversed theorem is in `CollatzResearch.RealTwoCoordinate`; it uses only `Affine.Nonnegative` and the eleven reversed weak comparisons.

Ambient types and the coefficient order. `formal/ReversedBinaryPowerClosure.lean`, lines 10–17.

```

10 variable {n : Type*} [Fintype n]
11
12 abbrev Vec (n : Type*) := n → ℝ
13 abbrev Mat (n : Type*) := Matrix n n ℝ
14
15 def EntrywiseLE (M N : Mat n) : Prop := ∀ i j, M i j ≤ N i j
16
17 local infix:50 " ≤ₘ " => EntrywiseLE

```

Affine composition and comparison. `formal/ReversedBinaryPowerClosure.lean`, lines 255–266.

```

255 structure Affine (n : Type*) where
256   matrix : Mat n
257   offset : Vec n
258
259 def Affine.comp (X Y : Affine n) : Affine n :=
260   (X.matrix * Y.matrix, X.matrix *ₘ Y.offset + X.offset)
261
262 def Affine.Nonnegative (X : Affine n) : Prop :=
263   (0 ≤ₘ X.matrix) ∧ 0 ≤ X.offset
264
265 def Affine.Weak (X Y : Affine n) : Prop :=
266   (Y.matrix ≤ₘ X.matrix) ∧ Y.offset ≤ X.offset

```

Two coordinates and admissibility. `formal/FullTwoBasic.lean`, lines 9–11.

```

9 abbrev Aff2 := Affine (Fin 2)
10
11 def Admissible (X : Aff2) : Prop := X.Nonnegative ∧ 1 ≤ X.matrix 0 0

```

Forward weak rules. `formal/FullTwoBasic.lean`, lines 21–32.

```

21 structure ForwardWeak (A B C D E F G : Aff2) : Prop where
22   ad : (A.comp D).Weak D
23   bd : (B.comp D).Weak (G.comp D)
24   ae : (A.comp E).Weak (E.comp A)
25   af : (A.comp F).Weak (E.comp B)
26   ag : (A.comp G).Weak (F.comp A)
27   be : (B.comp E).Weak (F.comp B)
28   bf : (B.comp F).Weak (G.comp A)
29   bg : (B.comp G).Weak (G.comp B)
30   ce : (C.comp E).Weak (C.comp B)
31   cf : (C.comp F).Weak (C.comp (A.comp A))
32   cg : (C.comp G).Weak (C.comp (A.comp B))

```

Forward offset equalities. `formal/FullTwoBasic.lean`, lines 34–45.

```

34 structure ForwardGapsZero (A B C D E F G : Aff2) : Prop where
35   ad : (A.comp D).offset 0 = D.offset 0
36   bd : (B.comp D).offset 0 = (G.comp D).offset 0
37   ae : (A.comp E).offset 0 = (E.comp A).offset 0
38   af : (A.comp F).offset 0 = (E.comp B).offset 0
39   ag : (A.comp G).offset 0 = (F.comp A).offset 0
40   be : (B.comp E).offset 0 = (F.comp B).offset 0
41   bf : (B.comp F).offset 0 = (G.comp A).offset 0
42   bg : (B.comp G).offset 0 = (G.comp B).offset 0
43   ce : (C.comp E).offset 0 = (C.comp B).offset 0
44   cf : (C.comp F).offset 0 = (C.comp (A.comp A)).offset 0
45   cg : (C.comp G).offset 0 = (C.comp (A.comp B)).offset 0

```

Reversed weak rules. Here the index type is finite and has decidable equality; the final theorem instantiates it with `Fin 2`. `formal/ReversedRealNormalization.lean`, lines 83–94.

```

83 structure ReversedRealWeak (A B C D E F G : Affine 1) : Prop where
84   da : (D.comp A).Weak D
85   db : (D.comp B).Weak (D.comp G)
86   ea : (E.comp A).Weak (A.comp E)
87   fa : (F.comp A).Weak (B.comp E)
88   ga : (G.comp A).Weak (A.comp F)
89   eb : (E.comp B).Weak (B.comp F)
90   fb : (F.comp B).Weak (A.comp G)
91   gb : (G.comp B).Weak (B.comp G)
92   ec : (E.comp C).Weak (B.comp C)
93   fc : (F.comp C).Weak (A.comp (A.comp C))
94   gc : (G.comp C).Weak (B.comp (A.comp C))

```

Reversed offset equalities. `formal/FullTwoBasic.lean`, lines 47–58.

```

47 structure ReversedGapsZero (A B C D E F G : Aff2) : Prop where
48   da : (D.comp A).offset 0 = D.offset 0
49   db : (D.comp B).offset 0 = (D.comp G).offset 0
50   ea : (E.comp A).offset 0 = (A.comp E).offset 0
51   fa : (F.comp A).offset 0 = (B.comp E).offset 0
52   ga : (G.comp A).offset 0 = (A.comp F).offset 0
53   eb : (E.comp B).offset 0 = (B.comp F).offset 0
54   fb : (F.comp B).offset 0 = (A.comp G).offset 0
55   gb : (G.comp B).offset 0 = (B.comp G).offset 0
56   ec : (E.comp C).offset 0 = (B.comp C).offset 0
57   fc : (F.comp C).offset 0 = (A.comp (A.comp C)).offset 0
58   gc : (G.comp C).offset 0 = (B.comp (A.comp C)).offset 0

```

The theorem with diagonal bounds. `formal/FullTwoCoordinate.lean`, lines 30–34.

```

30 theorem full_two_coordinate_obstruction (A B C D E F G : Aff2)
31   (hA : Admissible A) (hB : Admissible B) (hC : Admissible C) (hD : Admissible D)
32   (hE : Admissible E) (hF : Admissible F) (hG : Admissible G) :
33   (ForwardWeak A B C D E F G → ForwardGapsZero A B C D E F G) ∧
34   (ReversedRealWeak A B C D E F G → ReversedGapsZero A B C D E F G)

```

The unrestricted reversed theorem. `formal/CollatzReversedRealTwoCoordinate.lean`, lines 30–35.

```

30 theorem reversed_eligible_offsets_equal
31   (A B C D E F G : Affine (Fin 2))
32   (hA : A.Nonnegative) (hB : B.Nonnegative) (hC : C.Nonnegative)
33   (hD : D.Nonnegative) (hE : E.Nonnegative) (hF : F.Nonnegative) (hG : G.Nonnegative)
34   (h : ReversedRealWeak A B C D E F G) :
35   (D.comp A).offset = D.offset ∧ (D.comp B).offset = (D.comp G).offset

```